

Global Facilities - Design & Construction Standard - Construction - Completion Handover Business Process

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[ECN History](#)[Submit a Change Request](#)

1.0 Purpose

- 1.1 The purpose of this document is to provide the guidelines and instructions to execute the construction project Completion Handover Business Process that shall be followed and complied with during the execution and implementation of construction project handover, as part of any green and/or brown field construction projects contract requirement.

2.0 Scope

- 2.1 The guidelines in this document apply to any green and/or brown field construction projects and shall be utilized to conclude project completion handover activities by Micron Facilities team members, contractors, and vendors at any Micron construction project sites.
- 2.2 Please refer to [Global EHS – Construction and Design Performance Standard](#) and Global EHS Design Performance Standard for further guidance.

3.0 Definitions and Acronyms

3.1 Definitions of acronyms used in this document are listed as follows:

Acronym	Definition / Description
CAT	Category of defects
CAT A	Category of defects A: Must be completed before critical testing. Causing a safety concern (Example: Hydrotest. No more welding/cutting on the pressure envelope on the test pack). Prevents the commissioning to run in manual mode.
CAT B	Category of defects B: Must be completed before commissioning of the system to run in auto mode.
CAT C	Category of defects C: Must be completed before final contractor demobilization.
DCC	Document Control Center
FAT	Factory Acceptance Test
FCT	Facilities Central Team. This is used interchangeably with Global Facilities Engineering Team and associated to O&M BKM
GCT	Global Construction Team
GFTT	Global Facilities Technology Team. This is used interchangeably with Global Facilities Engineering Team and associated to design standards
GFC	Global Facilities & Construction Team. This is used interchangeably with Global Facilities (GFac), Facilities Central Group (FCG), World Wide Operations (WWOps), World Wide Facilities (WW_FACILITIES), Corp Facilities
Global Facilities Central Team	This is used interchangeably with Innovation & Integration Team
Global Facilities Construction Team	This is used interchangeably with Global Construction Team
Global Facilities Operations Support Team	This is used interchangeably with Global Facilities Operations Team
Global Facilities Planning Team	This is used interchangeably with Global Facilities Line Planning Team
MCCD	Mechanical and Commissioning Completion Dossier
NCR	Non-Conformance Report

Acronym	Definition / Description
OMM	Operation and Maintenance Manual
OPC	OLE Process Control
QA	Quality Assurance
QC	Quality Control
QOR	Quality Observation Report
RFE	Ready for Equipment
RFP	Request for Proposal
SAT	Site Acceptance Test
SCADA	Supervisory Control and Data Acquisition

4.0 Roles and Responsibilities

Roles	Responsibilities
GCT (Global Construction Project Management Teams)	Manage and lead team on the handover workflow activities of any project, which includes the contractor's handover proposal review and acceptance Ensure all execution are completed as per contract requirement, subsequently manage and support as appropriate with site project construction & operation teams, contractors and vendors Ensure related handover procedure, activities and documentation are included, as appropriate, to review the contractor's proposal to meet all related Micron standard, local regulation, requirement, code and standard Manage and track change management with site project construction team
Site Project Construction Teams	Provide technical feedback on Systems Handover standard and contractor's proposal Review and acceptance on contractor's proposal Incorporate systems handover into project design, engineering and construction Manage and track change management Document project deviation, proposal, expectation and any exclusion from Systems Handover Standard Ensure to capture in future version of Systems Handover Standard
Site Project Operation Teams	Collaborate with construction team to perform Systems Training, Test and Commissioning, operation, quality assurance and as-built documentation to achieve Handover completion Execute the Handover inspection until punches improvement is in acceptance condition Receive the Systems and Judge by Handover list Take over the Systems' operation when Handover completion

Roles	Responsibilities
FCT	Support site project operation team to perform Systems Test and Commissioning, operation, quality assurance and as-built documentation to achieve Handover completion Support site project operation team to ensure the Handover inspection until punches improvement is in acceptance condition Support site project operation team to receive the Systems and Judge by Handover list
Site Project EHS Team	Provide supervision and ensure effective implementation of MICRON Design, Construction and Handover of EHS Standards at site level and in compliance with local applicable Governmental Regulations Assist GCT in review of contractor's proposal to meet or comply with all related MICRON Handover Standards, local Regulations, requirements, local Codes and Standards Ensure that potential Contractors are appropriately briefed and communicated on MICRON Global EHS requirements and act as SME and technical expertise on EHS related design specifications
Procurement Teams	Review contractor's proposal on cost Validate cost of change and change management Ensure contractor's proposal must comply and to meet all related Micron standard, local regulation, requirement, code and standard
GFTT Lead (Discipline Engineer)	Support the Site team by providing knowledge and technical expertise as appropriate Manage and track change management with GCT and site project construction & Operation team

Note:

- 1) Statement highlighted in **YELLOW** – maintain this statement while removing the rest if this standard is to be communicated to Consultants/Contractors.
- 2) Statement highlighted in **GREY** – it is intended for internal reference and not to be shared with 3rd parties outside Micron.

5.0 Overview

5.1 Commencement of Completion Handover Business Process

The General and subcontractor are required to perform site inspection to ensure CAT A & B defects are identified and closed before project handover]

The systems and related equipment are required to run continuously in auto mode for a specified duration as outlined in the Contract and Micron specifications for each system/discipline.

The Project Construction Team, which includes the General Contractor, the Subcontractor and GCT, shall establish project RFP and relevant specifications as a basis for reference for the Project Operation Team. These documents shall be used during the project completion kick-off meeting to be held between both the teams in accordance with "Handover Alignment Checklist and Procedure".

5.2 Startup

The Project Construction Team shall prepare relevant testing plans, commissioning schedule and documents. The test plans, commissioning schedule and documents shall be categorized according to respective equipment's and systems. Once the documents are ready, the General Contractor shall submit a request to Micron to start equipment and system testing and commissioning. The GCT in collaboration with Site Project Operation Team shall review the submission. The General Contractor shall start the testing and commissioning only after addressing the comments raised during the review and after obtaining approval from Micron.

5.3 Completion Handover Procedure

The project handover procedure shall comprise Handover Preparation, Construction Completion, Quality Assurance, Handover Completion Documentation and Final Total Systems Acceptance.

5.4 Handover Preparation

As new construction project handover preparation readiness, the six indicators (5.4.1~5.4.6), and the following shall be made available by the Project Construction Team for handover to the Site Project Operation team:

- Utilities system testing and commissioning plan,
- System performance test,
- Systems in auto control mode,
- Training materials,
- Equipment and system manuals,
- System/equipment design calculations,
- Layout,
- Defects list,
- Mass balance from system to distribution,
- Calibration certificates,
- As-built drawings and documents, and
- Spare parts list, Vendor and Supplier contact

- 5.4.1 **Test and Commissioning:** The Contractor shall prepare an EHS Plan to specifically address environmental impact as well as health and safety risks arising from the testing and commissioning activities at least 3 months prior to commencement of Testing and Commissioning activities. The plan shall be presented to Micron for review and comments. The Contractor is required to apply Lock-out Tag-out on every system and subsystem to prevent unauthorized activation of hazardous energy. The Lock-out Tag out shall be removed to facilitate testing and commissioning through the permit-to-work system and reinstated immediately after completion of testing and commissioning. Field Management Enhancement protocol shall be adhered to strictly for testing and commissioning activities.
- The Project Construction Team to prepare and execute testing and commissioning for every single equipment and system. QA/QC shall be included as part of the testing and commissioning process for incoming/outgoing resource/waste e.g., domestic water, power, raw material, chemical, gas, sludge, etc. Every system shall undergo either a dynamic (test run, performance test, limit test, capacity test, valve test, etc.) or static test (I/O test, hydraulic capacity test, pressure test, pneumatic pressure test, power safety test, etc.) as deemed appropriate. The testing and commissioning shall be meet the acceptance criteria established by Micron.
- 5.4.2 **Functional Test (SAT):** The Project Construction Team shall plan and execute functional test for every single equipment. Testing of safety critical interlocks shall form part of the test. Functional test, failure test, redundancy test, control logic test, operational test and instrumentation test shall be carried out to ascertain equipment operational readiness.
- 5.4.3 **System in Auto Mode Control:** The Project Construction Team shall perform auto mode operation test for each system, which is inclusive of Zero/Full load test, peer to peer test, remote test, PLC/PID control test, timer test, etc., The auto mode operations test shall conform to Micron's operations team acceptance criteria.
- 5.4.4 **Training & Material:** The Project Construction Team shall make necessary arrangements to train and certify the Operations Team prior to handing over the new installed systems and equipment. The training shall include hazard communication, system design specifications, system operating procedure, system troubleshooting, SCADA and logic control, and other relevant training.
- 5.4.5 **Calibration & Records:** The Project Construction Team shall gather equipment and system calibration test certificates and register, which incorporates calibration frequencies, recommended service providers, equipment specifications, the need for regulatory approval, etc. The calibration certificates and related records shall

be made available and validated by Project Operation team during handover.

- 5.4.6 **As Built Documentation:** The Project Construction Team shall prepare all construction as-built drawings (PID, Layout, BIM, 3D Model in Autodesk Revit and Navisworks files format(.nwd), etc.), T&C reports, test records and all related documents (Punch List, SAT, OMM, MCCD, Document Control List, etc.), according to "As-built checklist documents as **Section 8.0** for handing over to Project Operation Team.

5.5 Construction Completion

- 5.5.1 **Auto Mode Control:** The utility systems are considered to be in auto control mode when the mechanical component and processes are stable with parameters correctly displayed on the FMCS online monitoring system. The operations in auto control mode shall be demonstrated by the Project Construction Team and accepted by Operations Team if acceptance criteria are met.

- 5.5.2 **FMCS Online:** In addition to the basic SCADA functions, SCADA alarm setting shall be followed BKM documents, OPC (data access/unified architecture) and all existing OPC connection to be made known and documented during handover, data historian servers must be set to enable big data extract, alarm history and operational abnormalities shall be recorded and available for user to extract and perform analysis. The system shall enable user defined report extraction as well.

5.6 Quality and EHS Assurance

- 5.6.1 **Inspection:** The Project Construction team is required to perform a preliminary inspection with the contractors before organizing a formal inspection/system walkdown with all the stakeholders, which includes Operation Team and EHS team. The findings from the inspection/system walkdown shall be transferred into a punch list, categorized according to severity (NCR, QOR, EHS, etc.) and tracked to closure. CAT A and CAT B rated findings shall be closed before starting-up or commissioning an equipment or performing SATs.

- 5.6.2 **Site Acceptance Test & Report:** Site Acceptance Test shall include the following scenarios: software FAT document, hydraulic validation, failure scenario test, black out test, major swing over for PLC, power test, full load test, etc. A report shall follow the test performed. CAT A and CAT B rated findings shall be closed.

- 5.6.3 **Improvement & Confirmation:** All CAT A or CAT B quality and EHS related deviations in the punch list shall be rectified before the Project Construction Team, Operation Team, EHS team and other relevant stakeholders agree to system or equipment handover.

The contractor/vendor is allowed to perform system retuning, where required and if agreed by the stakeholders, after the

handover process if the project timeline does not allow for such activity.

5.7 Handover Completion

During the completion of “handover” inspections, the General Contractor / Subcontractor shall submit all related documents, which includes As-built drawings, documents, records, etc., to Micron Project Construction Team, Project Site DCC for safekeeping, and Government or Relevant Agency. The documents that need to hand over is not limited to but are as follows:

- 5.7.1 **Test and Commissioning Record:** Test and Commissioning reports, records and where applicable manufacturer’s certificates for the equipment, related instrument, and components.
- 5.7.2 **As Built Documents / Records:** RFP Document, As-Built Drawings, Layout Drawings, DCC check list, and Handover List.
- 5.7.3 **Spare Parts & Warranty:** Spare parts and consumable material lists, Contract warranty documents.
- 5.7.4 **System Handover:** SAT, Troubleshooting Description and Training Document, Operation and Maintenance Manual.
- 5.7.5 **Space Handover:** Meet Facility Operation Team, Area Owners, EHS, Security, ERC, IT and QA requirements.
- 5.7.6 **Environment Handover:** Lighting, Signage / Labeling, Clean up and Housekeeping Execution Plan.

5.8 Change management

If there is a need for additional changes during the “handover” process, those changes shall be in conformance to Micron’s CMOC (Construction Management of Change) process.

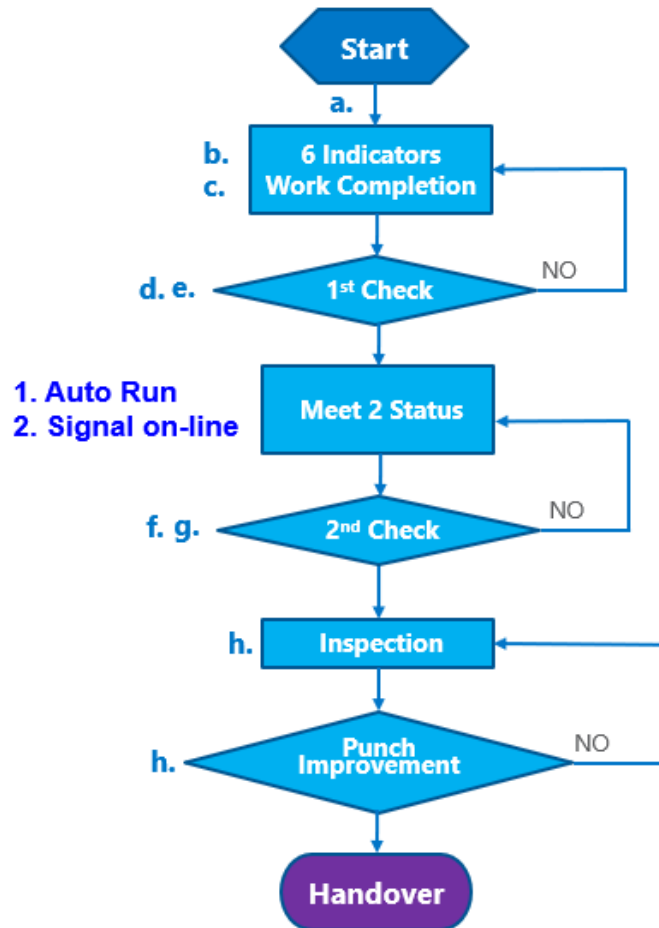
6.0 Handover Workflow:

The key activities listed below, and the flowcharts provided here are to be referred to by the Project Construction Team, which includes the Contractors.

6.1 Key activities:

- a) Handover List Alignment (Confirmed by Operation & Construction Teams)
- b) indicators work completion
- c) Training
- d) 1st As Built Document confirmation
- e) 1st Inspection & Improvement
- f) 2nd As Built Document confirmation
- g) 2nd Operation Training
- h) 2nd Inspection & Improvement

6.2 Activities Flow Chart (Sample for Reference):



7.0 Completion Handover Documentation

The handover completion documentation comprising of handover procedure, construction completion, quality assurance and handover completion (please refer to **Section 8** for details of handover documentations) shall be completed to allow for the building and utilities system handover transition from the Project Construction team to Project Operations Team.

8.0 As-Built Document Checklists (Final Documentation)

Please refer to **Table 1** for details of handover documents.

Table 1: Key items of As-built Document Checklists Summary Table

No.	Items	Description
A	Engineering Manual	RFP & Contract Documents, Organization Chart, Manpower Table, Total Completion Schedule, As-Built Progress, etc.
B	Equipment Specification	SPEC Data Sheet, System & Equipment SPEC, Pump Curve, System Curve & Function Calculation Sheet, etc.
C	Calculation Sheet	Design/Structure Calculation, System/Equipment Performance Calculation & Curve, etc.
D	System/ Equipment/ Material Test	System/Equipment/Material Test Report, Incoming Inspection Report, Factory Acceptance Test (FAT) Report, System/Equipment Performance Quality Verification Test Report, Construction Completion Test Report, etc.
E	Equipment/ Pipeline Anti-Vibration Design	Equipment/pipeline Anti-Earthquake & Micro-Vibration calculation Report/Test Record, etc.
F	FMCS	Monitoring and Control System Report, Process Flow Diagram, PID Diagram, I/O List, Control SOO, Control System Single Line Diagram, Instrument Test Report, SAT Report, etc.
G	Construction Engineering Change Management Report	SCR, CRI, VO & Cost Report, etc.
H	Test & Commissioning Report	System/Equipment Test & Operation Report & Records, Power Factor Records, System/Equipment Balance Test Records, System Acceptance Test (SAT) Report, etc.

No.	Items	Description
J	Certificate	Incoming Report, Factory Report, Factory Certificate, Occupancy Permit, Fire Fighting Permit, Sustainability LEED Certificate, Reclaim Acceptance Certification Performance % for Water & Reclaim / Waste System, etc.
K	Acceptance Records	Contractors Self-Inspection Check List, Initial Inspection Report, Re-inspection Report, Punch List, Piping & Tank Pressure Test Records, Equipment Control Point-to-Point Test Record, Piping Analytical Testing (oxygen, moisture, and particles), Genealogy of Supply and Drainage Pipelines, etc.
L	Factory/Original Certificate	Import Certificate, Sales Certificate, Factory Certificate, OEM Factory Test Report & Analytical Certification Test Report, etc.
M	Operation Manual	System/Equipment Operation Manual, Education and Training Course Materials and Test Papers, etc.
N	Brief Operation Manual	Main Operating Procedure, etc.
O	Maintenance Instructions and Spare Parts	System/Equipment/Parts Maintenance Manual & Cycle, Equipment/Parts Catalog, Equipment/Parts List and Supplier (Including Agent) Information, Recommended Spare Parts List, Handover Spare Parts List, Consumable Material List, etc.
P	Engineering Photos	Construction Photos, As Built Photos, etc.
Q	As Built Drawings	Single Line Diagram, Equipment/Piping Layout, Equipment Dimension Details, Equipment & Piping Arrangement Drawings, Section View Diagrams, Shop Drawings, etc.
R	Other Documents	Engineering Daily/Weekly Report, Others (Depending on System Characteristics or Sponsors Requirements)
S	EHS Report	Site Administration/EHS Check List, Facility Process and Supply System EHS Check List, EHS-Design for Safety Check List, etc.

The details of items for As-built Document Checklists are stipulated in the Appendix Section within this document:

Appendix-1: "Facilities MEP System As-Built Acceptance Document Checklist"

Appendix-2: "Facilities CSA System As-Built Acceptance Document Checklist"

Appendix-3: "BIM Model As-Built Acceptance Document Checklist"

Appendix-4: "Facilities Process and Supply System EHS Checklist" (Template Sample for Reference)

Appendix-5: "EHS-Design in Safety Checklist and Maintainability Guide" (Must be Maintainability Check and SME Validation).

Appendix-6: Project Handover Checklist & Client Acceptance Document (Template Sample for Reference)

Note:

- 1) **"Appendix-4"** - The purpose of this check list is to ensure that each completed system was being evaluated and confirmed on its compliance with EHS requirement prior to start-up. Facility should utilize this checklist to complete the inspection for EHS review and further confirmation.
- 2) **"Appendix-5"** - The purpose of this check list is to ensure that "SID (Safety in Design)" program has been performed "Safe Design Methodology" of "Global EHS - Design Performance Standard" when the document was approved and launched for implementation.

9.0 Final Total Systems Acceptance:

After Handover Business Process has been completed, General Contractor shall collect all documents and records which are listed and signed off by relevant persons (Template Sample of "Project Handover Checklist & Client Acceptance Document" Please refer to **Appendix 6**), then submit to MICRON Project site DCC for Project Construction and Operation Teams approval and confirmation. When both MICRON construction and operation teams are agreed and accepted this "Completion Handover Documentation", it will be called at the end of Handover Completion as achievement of "Final Total Systems Acceptance".

10.0 Document Control

10.1 Approval

FCG_ENG_CONSTRUCTION_APPR

10.2 Notification

GLOBAL_EHS
GLOBAL_EHS_SAFETYHEALTH
GLOBAL_FAC_COMMUNICATIONS
GLOBAL_FAC_CONST_SOP
GLOBAL_FAC_MANAGERS
GLOBAL_FAC_NOTIFY
GLOBAL_SECURITY_TEAM
GP_INDIRECT_FACILITIES
FCG_F4_FAC_NOTIFY
FCG_F6_FAC_NOTIFY
FCG_F10_FAC_NOTIFY
FCG_F11_FAC_NOTIFY
FCG_F15_FAC_NOTIFY
FCG_F16_FAC_NOTIFY
FCG_MMP_FAC_NOTIFY
FCG_MMY_FAC_NOTIFY
FCG_MSB_FAC_NOTIFY
FCG_MTB_FAC_NOTIFY
FCG_MXA_FAC_NOTIFY
WW_FACILITIES_MANAGERS_APPR

10.3 Final Viewer Access

All Micron sites Facilities Team Members

10.4 Retention

Adherence to the requirements specified in the
[**Global Facilities - Document Control Procedures**](#)

10.5 Review

Biennial

11.0 Revision History

Revision #	Section Changed	Details of Changes	Date
0	NA	New document	09/20/21

Appendix-1.0: Facilities MEP System As-Built Acceptance Documents Checklist**No.: A – 01**

Date : Year Month Day

No.	Items	Result	Remark
A.	Engineering Manual	☐ Yes ☐ No ☐ NA	
1.	Manpower Table	☐ Yes ☐ No ☐ NA	
2.	Total Completion Schedule	☐ Yes ☐ No ☐ NA	
B.	List of system/equipment specifications	☐ Yes ☐ No ☐ NA	
1.	System Specification List	☐ Yes ☐ No ☐ NA	
2.	Equipment Specification List	☐ Yes ☐ No ☐ NA	
3.	Comparison table of equipment specifications	☐ Yes ☐ No ☐ NA	
C.	Design/Structure Calculation (Include PE endorsements if the document is required.)	☐ Yes ☐ No ☐ NA	All the respective pipe/utilities rack inside and outside the buildings shall be included. Site/Building/Room Boundary noise shall be included.
1.	System Performance Calculation	☐ Yes ☐ No ☐ NA	System final quality
2.	Equipment Performance Calculation	☐ Yes ☐ No ☐ NA	Individual equipment performance quality.
3.	Equipment Performance Curve	☐ Yes ☐ No ☐ NA	80% of equipment capacity
4.	Pumps/Windmills in Parallel/Series Curve	☐ Yes ☐ No ☐ NA	
5.	Piping Network Design Calculation with appropriate software used	☐ Yes ☐ No ☐ NA	
6.	Electrical original software, driver, communication tool and Accompanying Manual Information (password, etc.)	☐ Yes ☐ No ☐ NA	
7.	List of Cable schedule	☐ Yes ☐ No ☐ NA	

No.	Items	Result	Remark
8.	Electrical system studies report (software files)	☐ Yes ☐ No ☐ NA	
D.	System/Equipment/Material Test Report	☐ Yes ☐ No ☐ NA	
1.	System/Equipment/Material Incoming Inspection Report	☐ Yes ☐ No ☐ NA	
2.	System/Equipment/Material Factory Acceptance Test (FAT) Report	☐ Yes ☐ No ☐ NA	
3.	Construction Completion Test Record Report	☐ Yes ☐ No ☐ NA	
4.	System/Equipment/Material Performance Quality Verification Test Report	☐ Yes ☐ No ☐ NA	Quality inspection report shall be included, such as pipe threading & welding joint inspection, pressure test, electrical cable ladder loading test report, etc.
E.	Equipment/Pipeline Anti-Earthquake / Micro Vibration Design	☐ Yes ☐ No ☐ NA	For example: the calculation of mechanical equipment/pipeline, electrical cable ladder support, etc.
1.	Equipment/pipeline 0.65G Anti-Vibration calculation sheet	☐ Yes ☐ No ☐ NA	For example: the calculation of mechanical equipment/pipeline, electrical cable ladder support, etc.
2.	Equipment/Piping Micro Vibration Calculation Sheet	☐ Yes ☐ No ☐ NA	
3.	Micro-vibration three-stage Check list (specification 15840)	☐ Yes ☐ No ☐ NA	
4.	Support As-Built Drawing	☐ Yes ☐ No ☐ NA	
5.	Anti-Vibration Inspection Record	☐ Yes ☐ No ☐ NA	
F.	Monitoring and Safety-Related System Acceptance	☐ Yes ☐ No ☐ NA	Include all the failure scenario tested, black out, restart, major swing over for PLC & Power Tested) with Report and Open Item Closed. All Critical System Must have N+1 Redundancy at the time of handover.
1.	Piping & Instrument Diagram (P&ID)	☐ Yes ☐ No ☐ NA	
2.	Control System Architecture	☐ Yes ☐ No ☐ NA	

No.	Items	Result	Remark
3.	Instrument Control Equipment & Piping Layout Plan	☐ Yes ☐ No ☐ NA	
4.	Instrument Control Equipment Riser Diagram	☐ Yes ☐ No ☐ NA	
5.	Instrument Control Equipment Cable Tray Layout Plan	☐ Yes ☐ No ☐ NA	
6.	Instrument Control Equipment Signal Cable Route Diagram	☐ Yes ☐ No ☐ NA	
7.	Device Network & Field Bus Diagram of Instrument Control System (e.g., DeviceNet, Profibus Modbus, optical fiber)	☐ Yes ☐ No ☐ NA	
8.	Control Power Distribution Diagram	☐ Yes ☐ No ☐ NA	
9.	Detailed Grounding Diagram of the I&C System	☐ Yes ☐ No ☐ NA	
10.	Tag Name (Naming Principle)	☐ Yes ☐ No ☐ NA	
11.	I/O List (Assignment) and Point Statistics Table	☐ Yes ☐ No ☐ NA	
12.	CPU and Remote I/O control panel design document	☐ Yes ☐ No ☐ NA	
13.	Control Sequence of Operations (SOO)	☐ Yes ☐ No ☐ NA	
14.	Single Line of Control System Diagram	☐ Yes ☐ No ☐ NA	
15.	Instrument Control Equipment Factory Test Report	☐ Yes ☐ No ☐ NA	
16.	Instrument Control Equipment Operation Manual	☐ Yes ☐ No ☐ NA	
17.	Instrument Control Equipment Maintenance Manual (Including Maintenance Cycle)	☐ Yes ☐ No ☐ NA	

No.	Items	Result	Remark
18.	Instrument Control Equipment Catalog	☐ Yes ☐ No ☐ NA	
19.	List of Instrument Control Equipment and Information of Suppliers (Including Agents)	☐ Yes ☐ No ☐ NA	
20.	Calibration Record of Instrument Control Equipment	☐ Yes ☐ No ☐ NA	
21.	Instrument Control Equipment Parameter Data	☐ Yes ☐ No ☐ NA	
22.	Monitoring System Operation and Maintenance Manual	☐ Yes ☐ No ☐ NA	
23.	Original Software and Accompanying Manual Information (include PLC Logic with Source Code)	☐ Yes ☐ No ☐ NA	
24.	Monitoring Program Image File	☐ Yes ☐ No ☐ NA	
25.	PFD (Process Flow Diagram) System Introduction and System Flow	☐ Yes ☐ No ☐ NA	
26.	SCADA / FMCS Implementation of the Initial Inspection Report	☐ Yes ☐ No ☐ NA	
27.	Factory Acceptance Test Report (FAT Report)	☐ Yes ☐ No ☐ NA	
28.	Site Acceptance Test Report (SAT Report)	☐ Yes ☐ No ☐ NA	Include Full Load with Report and Open Item Closed.
29.	Device Network & Field Bus (e.g., DeviceNet, Profitbus, Modbus) Test Report	☐ Yes ☐ No ☐ NA	
30.	Test Report of the Grounding Circuit of the Instrument Control System	☐ Yes ☐ No ☐ NA	
31.	A List of Spare Parts for the Monitoring System (A list of spare parts with the same specifications from the original manufacturer must be provided)	☐ Yes ☐ No ☐ NA	
32.	Exhaust Test & Balance Report		Include Exhaust test & balance for gas cabinets and VMB. Provide the CFD (Computer Fluid Dynamic) simulation report, if the system is requested.
G.	Construction Engineering Change Management Report (SCR, CRI, VO & Cost Report, etc.)	☐ Yes ☐ No ☐ NA	
H.	Test & Commissioning Report	☐ Yes ☐ No ☐ NA	

No.	Items	Result	Remark
1.	System/Equipment Test & Operation Records	☐ Yes ☐ No ☐ NA	Include Final List of Parameter Setting for VFD/VSD, etc.
2.	Piping/Line Test Records before System/Equipment Test Operation	☐ Yes ☐ No ☐ NA	
3.	Electrical Test Records before System/Equipment Test Operation	☐ Yes ☐ No ☐ NA	
4.	System/Equipment Supply Quality Verification Records	☐ Yes ☐ No ☐ NA	
5.	Air Volume, Water Volume & Mass Balance Test Records	☐ Yes ☐ No ☐ NA	
6.	Equipment Power Factor Validation Records	☐ Yes ☐ No ☐ NA	
J.	Permit / Certificate	☐ Yes ☐ No ☐ NA	Include tanks, pressure vessel, etc.
1.	Occupancy Permit	☐ Yes ☐ No ☐ NA	
2.	Fire Fighting System Permit	☐ Yes ☐ No ☐ NA	
3.	LEED Certificate	☐ Yes ☐ No ☐ NA	
4.	Reclaim Acceptance Certification Performance % for Water & Reclaim / Waste System.	☐ Yes ☐ No ☐ NA	System design criteria and specification including calculation have to be shown in handover document.
K.	Acceptance Records	☐ Yes ☐ No ☐ NA	
1.	Contractors Self-inspection Check List	☐ Yes ☐ No ☐ NA	
2.	Initial Inspection Record	☐ Yes ☐ No ☐ NA	
3.	Re-inspection Record	☐ Yes ☐ No ☐ NA	
4.	Punch List	☐ Yes ☐ No ☐ NA	
5.	Piping and Tank Pressure Test Records	☐ Yes ☐ No ☐ NA	
6.	Equipment Control Point-to-Point Test Record	☐ Yes ☐ No ☐ NA	

No.	Items	Result	Remark
7.	Genealogy of Supply and Drainage Pipelines	☐ Yes ☐ No ☐ NA	
8.	Performance Test Report	☐ Yes ☐ No ☐ NA	
L.	Factory/Original Certificate	☐ Yes ☐ No ☐ NA	
1.	Import Certificate	☐ Yes ☐ No ☐ NA	
2.	Sales Certificate	☐ Yes ☐ No ☐ NA	
3.	Factory Certificate	☐ Yes ☐ No ☐ NA	
4.	OEM Factory Test Report & Analytical Certification Test Report	☐ Yes ☐ No ☐ NA	
M.	Instructions	☐ Yes ☐ No ☐ NA	
1.	System Operation Manual (System Operating Instruction)	☐ Yes ☐ No ☐ NA	Include Ops 10 years operation and Caps equipment price (TCOO).
2.	Equipment Operation Manual	☐ Yes ☐ No ☐ NA	
3.	Education and Training Course Materials and Test Papers	☐ Yes ☐ No ☐ NA	
N.	Brief Operation Manual	☐ Yes ☐ No ☐ NA	
O.	Maintenance Instructions and Spare Parts (Include the information of manufacturer/vendors and contact)	☐ Yes ☐ No ☐ NA	
1.	System/Equipment/Parts Maintenance Instruction	☐ Yes ☐ No ☐ NA	
2.	System/Equipment/Parts Maintenance Cycle	☐ Yes ☐ No ☐ NA	
3.	Equipment/Parts Catalog	☐ Yes ☐ No ☐ NA	
4.	Equipment/Parts List and Supplier (Including Agent) Information	☐ Yes ☐ No ☐ NA	Include the maintenance interval and key parts replacement interval.

No.	Items	Result	Remark
5.	Recommended Spare Parts List	☐ Yes ☐ No ☐ NA	
6.	Handover Spare Parts List	☐ Yes ☐ No ☐ NA	
P.	Engineering Photos	☐ Yes ☐ No ☐ NA	
1.	Construction Photos	☐ Yes ☐ No ☐ NA	
2.	As-Built Photos	☐ Yes ☐ No ☐ NA	
Q.	As-Built Drawings (Final Revised Version) (CAD Files and Paper Copies Are Provided)	☐ Yes ☐ No ☐ NA	
1.	Single Line Diagram of Each System Piping / Ducting, etc.	☐ Yes ☐ No ☐ NA	
2.	Layout of Each System Pipeline	☐ Yes ☐ No ☐ NA	
3.	Layout of Equipment Power Panel	☐ Yes ☐ No ☐ NA	
4.	Power system Single Line Diagram	☐ Yes ☐ No ☐ NA	
5.	Electric Cable/Tray Layout	☐ Yes ☐ No ☐ NA	
6.	Equipment Dimension Details	☐ Yes ☐ No ☐ NA	
7.	Equipment Layout Drawings	☐ Yes ☐ No ☐ NA	
8.	Equipment Electrical and Pipeline Diagram, Wiring Duct Distribution Diagram	☐ Yes ☐ No ☐ NA	
9.	Piping Arrangement Drawings	☐ Yes ☐ No ☐ NA	
10.	Typical Shop Drawings (Details)	☐ Yes ☐ No ☐ NA	
11.	Section View Drawings (Riser Flow Diagram)	☐ Yes ☐ No ☐ NA	
12.	Layout & Section View of Space Management Drawings	☐ Yes ☐ No ☐ NA	

No.	Items	Result	Remark		
R.	Other Documents	☐ Yes ☐ No ☐ NA			
1.	Engineering Daily Report	☐ Yes ☐ No ☐ NA			
2.	Engineering Weekly Report (Only the Original Electronic File Is Required)	☐ Yes ☐ No ☐ NA			
3.	FMEA & HAZOP system	☐ Yes ☐ No ☐ NA			
4.	Others (Depending on System Characteristics or Sponsors Requirements)	☐ Yes ☐ No ☐ NA			
Conclusion:					
Site Project Construction Direct Manager	Site Project Operation Direct Manager	Site Project Construction System Sponsor	Site Project Operation System Sponsor	GCT Engineering Management	Contractor

No.: B – 01

Date: Year Month Day

No.	Check Items	Result	Remark
1.	All equipment and articles borrowed from the owner have been returned.	☐ Yes ☐ No ☐ NA	
2.	All temporary construction facilities, construction waste, and garbage have been removed in accordance with the requirement.	☐ Yes ☐ No ☐ NA	
3.	All public facilities damaged in accordance with the construction have been restored.	☐ Yes ☐ No ☐ NA	
4.	Temporary power/pipelines have been removed completely.	☐ Yes ☐ No ☐ NA	
5.	Hazard labeling (SDS data sheet and GHS labeling).	☐ Yes ☐ No ☐ NA	
6.	All fines for violations of safety, health and environment protection have been settled.	☐ Yes ☐ No ☐ NA	
7.	Temporary utility fees e.g., water/electricity payables have been fully settled.	☐ Yes ☐ No ☐ NA	
8.	Verification of dangerous workplace has been acknowledged by the Ministry of Labor.	☐ Yes ☐ No ☐ NA	
9.	Design has complied with relevant environmental and health regulations.	☐ Yes ☐ No ☐ NA	
10	Construction, installation or testing are in compliance with safety design requirements.	☐ Yes ☐ No ☐ NA	

No.	Check Items	Result	Remark
11.	Affected existing control/safety facilities have been completely corrected and adjusted to maintain operational conditions.	☐ Yes ☐ No ☐ NA	
12.	Complied with Micron's safety protection specifications (Ex: high rise, CoHE, electric, confined space).	☐ Yes ☐ No ☐ NA	
13.	All emergency exits and evacuation routes remain unblocked.	☐ Yes ☐ No ☐ NA	
14.	Installation of permanent electrical equipment (switches, circuit breakers, wiring, anti-electric leakage sockets, etc.) are in compliance with regulation.	☐ Yes ☐ No ☐ NA	
15.	Floor openings include trench covers, inspection holes, and pits are properly covered.	☐ Yes ☐ No ☐ NA	
16.	The lighting of all places in the building meets the requirements of codes and regulations.	☐ Yes ☐ No ☐ NA	
17.	Safety instructions of safety equipment and factory original inspection certificate are provided.	☐ Yes ☐ No ☐ NA	

Conclusion:

Site Project Construction Direct Manager	Site Project Operation Direct Manager	Site Project Construction System Sponsor	Site Project Operation System Sponsor	GCT Engineering Management	Contractor

Appendix-2.0: Facilities CSA System As-Built Acceptance Document Checklist**No.: A – 02**

Date : Year Month Day

Item	Check Items	Check Result	Remark
A.	Construction Manual (Required)	☐ Yes ☐ No ☐ NA	
1.	Manpower Table (Required)	☐ Yes ☐ No ☐ NA	
2.	Total Completion Schedule (Required)	☐ Yes ☐ No ☐ NA	
B.	List of system/equipment specifications	☐ Yes ☐ No ☐ NA	
1.	Elevators	☐ Yes ☐ No ☐ NA	
2.	Automatic Door	☐ Yes ☐ No ☐ NA	
3.	Landscape Sprinkler System	☐ Yes ☐ No ☐ NA	
4.	Electric Metal Rolling Door	☐ Yes ☐ No ☐ NA	
5.	Electric Telescopic Door	☐ Yes ☐ No ☐ NA	
6.	Door, Wall & Window Machine (Fire rated / Electric)	☐ Yes ☐ No ☐ NA	
7.	Lifting Platform	☐ Yes ☐ No ☐ NA	
8.	Fence Machine	☐ Yes ☐ No ☐ NA	
9.	Other Electrical Equipment	☐ Yes ☐ No ☐ NA	
C.	Design/Structure Calculation (Include PE endorsements if the document is required.)	☐ Yes ☐ No ☐ NA	
1.	Floor Structure	☐ Yes ☐ No ☐ NA	Include design calculation, load chart.
2.	Curtain Wall	☐ Yes ☐ No ☐ NA	

Item	Check Items	Check Result	Remark
3.	Wall Window Machine (Electric)	☐ Yes ☐ No ☐ NA	
4.	Fire / Explosion Venting Wall / Door	☐ Yes ☐ No ☐ NA	
5.	Vent Wall	☐ Yes ☐ No ☐ NA	
6.	Ceiling Hanging Structure	☐ Yes ☐ No ☐ NA	Include PE endorsement and states max load for future AMHS.
D.	System/Equipment/Material Test Report	☐ Yes ☐ No ☐ NA	
1.	Equipment/Material Incoming Inspection Report	☐ Yes ☐ No ☐ NA	
2.	Equipment/Material Factory Test Report	☐ Yes ☐ No ☐ NA	
3.	Construction Completion Test Record Report	☐ Yes ☐ No ☐ NA	
4.	Equipment Performance Quality Verification Test Report	☐ Yes ☐ No ☐ NA	
G.	Construction Engineering Change Management Report (SCR, CRI, VO & Cost Report, etc.)	☐ Yes ☐ No ☐ NA	
H.	Test & Commissioning Report	☐ Yes ☐ No ☐ NA	
1.	Elevators	☐ Yes ☐ No ☐ NA	
2.	Automatic Door	☐ Yes ☐ No ☐ NA	
3.	Landscape Sprinkler System	☐ Yes ☐ No ☐ NA	
4.	Electric Metal Rolling Door	☐ Yes ☐ No ☐ NA	
5.	Electric Telescopic Door	☐ Yes ☐ No ☐ NA	

Item	Check Items	Check Result	Remark
6.	Wall Window Machine (Electric)	☐ Yes ☐ No ☐ NA	
7.	Lifting Platform	☐ Yes ☐ No ☐ NA	
8.	Fence Machine	☐ Yes ☐ No ☐ NA	
9.	Other Electrical Equipment	☐ Yes ☐ No ☐ NA	
J.	Permit / Certificate	☐ Yes ☐ No ☐ NA	
1.	Occupancy Permit	☐ Yes ☐ No ☐ NA	
2.	Fire Fighting System Permit	☐ Yes ☐ No ☐ NA	
3.	LEED Certificate	☐ Yes ☐ No ☐ NA	
K.	Acceptance Records (Required)	☐ Yes ☐ No ☐ NA	Include the listed items below but not limited: a. External storm drain pH checked due to ground water issue. b. All wet area floor, trench and sump pit to have water proofing membrane or water proofing coating or water proofing additives. c. Concrete kerb design on any floor penetration. d. External area floor flatness to avoid water ponding.
1.	Contractors Self-Inspection Check List	☐ Yes ☐ No ☐ NA	
2.	Initial Inspection Record	☐ Yes ☐ No ☐ NA	
3.	Re-inspection Record	☐ Yes ☐ No ☐ NA	
4.	Punch List	☐ Yes ☐ No ☐ NA	
L	Factory/Original Certificate	☐ Yes ☐ No ☐ NA	

Item	Check Items	Check Result	Remark
1.	Import Certificate	☐ Yes ☐ No ☐ NA	
2.	Sales Certificate	☐ Yes ☐ No ☐ NA	
3.	Factory Certificate	☐ Yes ☐ No ☐ NA	
M.	System Operation Manual	☐ Yes ☐ No ☐ NA	
1.	Water Pool Equipment	☐ Yes ☐ No ☐ NA	
2.	Landscape Sprinkler System	☐ Yes ☐ No ☐ NA	
3.	Elevators	☐ Yes ☐ No ☐ NA	
4.	Lifting Platform	☐ Yes ☐ No ☐ NA	
5.	Electric Metal Rolling Door	☐ Yes ☐ No ☐ NA	
6.	Electric Telescopic Door	☐ Yes ☐ No ☐ NA	
7.	Automatic Door	☐ Yes ☐ No ☐ NA	
8.	Window Machine (Electric)	☐ Yes ☐ No ☐ NA	
9.	Other Electrical Equipment	☐ Yes ☐ No ☐ NA	
N.	Brief Operation Manual	☐ Yes ☐ No ☐ NA	
1.	Landscape Sprinkler System	☐ Yes ☐ No ☐ NA	
2.	Lifting Platform	☐ Yes ☐ No ☐ NA	

Item	Check Items	Check Result	Remark
3.	Automatic Door	☐ Yes ☐ No ☐ NA	
4.	Other Electrical Equipment	☐ Yes ☐ No ☐ NA	
O.	Maintenance Instructions and Spare Parts List	☐ Yes ☐ No ☐ NA	
1.	Equipment/Parts Maintenance Manual	☐ Yes ☐ No ☐ NA	
2.	Equipment/Parts Maintenance Cycle	☐ Yes ☐ No ☐ NA	
3.	Equipment/Parts Catalog	☐ Yes ☐ No ☐ NA	
4.	Equipment/Parts List and Supplier (Including Agent) Information	☐ Yes ☐ No ☐ NA	
5.	Recommended Spare Parts List	☐ Yes ☐ No ☐ NA	
6.	Handover Spare Parts List	☐ Yes ☐ No ☐ NA	
P.	Engineering As-Built Photos	☐ Yes ☐ No ☐ NA	
1.	Construction Photos	☐ Yes ☐ No ☐ NA	
2.	As-Built Photos	☐ Yes ☐ No ☐ NA	
Q.	As-Built Drawings (Final Revised Version) (CAD Files and Paper Copies Are Provided) (Required)	☐ Yes ☐ No ☐ NA	
1.	Layout Drawings	☐ Yes ☐ No ☐ NA	
2.	Section Drawings	☐ Yes ☐ No ☐ NA	
3.	Cross Section Drawings	☐ Yes ☐ No ☐ NA	
4.	Punch List	☐ Yes ☐ No ☐ NA	

Item	Check Items	Check Result	Remark
5.	Typical Shop Drawings (Details)	☐ Yes ☐ No ☐ NA	
6.	Approved Opening Drawings	☐ Yes ☐ No ☐ NA	
R.	Other Documents	☐ Yes ☐ No ☐ NA	
1.	Engineering Daily Report (Required)	☐ Yes ☐ No ☐ NA	
2.	Engineering Weekly Report (Only the Original Electronic File Is Required)	☐ Yes ☐ No ☐ NA	
3.	Approved Submittals List	☐ Yes ☐ No ☐ NA	
4.	EHS Inspection Completed Affidavit	☐ Yes ☐ No ☐ NA	
5.	Others (Depending on System Characteristics or Sponsors Requirements)	☐ Yes ☐ No ☐ NA	

Conclusion:

Site Project Construction Direct Manager	Site Project Operation Direct Manager	Site Project Construction System Sponsor	Site Project Operation System Sponsor	GCT Engineering Management	Contractor

No.: B – 02

Date: Year Month Day

No.	Check Items	Result	Remarks		
1.	All equipment and articles borrowed from the owner have been returned.	☐ Yes ☐ No ☐ NA			
2.	All temporary construction facilities, construction waste, and garbage have been removed in accordance with the requirement.	☐ Yes ☐ No ☐ NA			
3.	All public facilities damaged in accordance with the construction have been restored.	☐ Yes ☐ No ☐ NA			
4.	Temporary power/pipelines have been removed completely.	☐ Yes ☐ No ☐ NA			
5.	Hazard labeling (SDS data sheet and GHS labeling).	☐ Yes ☐ No ☐ NA			
6.	All fines for violations of safety, health and environment protection have been settled.	☐ Yes ☐ No ☐ NA			
7.	Temporary utility fees e.g., water/electricity payables have been fully settled.	☐ Yes ☐ No ☐ NA			
Conclusion:					
Site Project Construction Direct Manager	Site Project Operation Direct Manager	Site Project Construction System Sponsor	Site Project Operation System Sponsor	GCT Engineering Management	Contractor

Appendix 3.0 BIM Model Acceptance Document Checklist**Appendix-3.1:** BIM Model Design Acceptance Document Checklist**No.: A – 3.1**

Date: Year Month Day

Project No/Name: _____

First Check Date: _____ Micron Personnel Name: _____

Vendor Personnel Name: _____

Final check Date: _____ Micron Personnel Name: _____

Model: _____ Vendor Personnel Name: _____

No.	Items	1st Check	Final Check	Reason/s
		✓ = OK, ✕ = Not OK		
1.0	3D model has been coordinated and clash resolved. (Include clash matrix, graph, clash rules sets, clash report xml and clash detection model files for review.)			
2.0	3D models modelled to construction model requirements. <i>(Please attach construction 3D model requirement checklist. Please state reason/s if not provided.)</i>			
3.0	The 2D drawings are direct produced from 3D.			
4.0	Maintenance area and clearance models included and acknowledged by FM O&M team. <i>(Note down comments, if any)</i>			
5A	P&ID, SL, Schematics, Riser, Systems diagrams match 3D model.			
5B	P&ID, SL, Schematics, Raiser, Systems diagrams tags are same in 3D model.			
Electrical				
5.1	List systems with P&ID, single line, schematic, flow, system, riser diagrams.			
Mechanical HVAC				

No.	Items	1st Check	Final Check	Reason/s
		✓ = OK, ✕ = Not OK		
5.2	List systems with P&ID, single line, schematic, flow, system, riser diagrams.			
6	CSA schedules are created for:			
6.1	Multi-Category			
6.2	Assemblies			
6.3	Railings			
6.4	Railings Handrails			
6.5	Railings Support			
6.6	Railings Top Rails			
6.7	Ramps			
6.8	Roofs			
6.9	Roofs Fascias			
6.10	Roofs Gutters			
6.11	Roofs Soffits			
6.12	Site Pads			
6.14	Stair Landings			
6.15	Stair Runs			
6.16	Stair Supports			
6.17	Topography			
6.18	Walls Wall Sweeps			

No.	Items	1st Check	Final Check	Reason/s
		✓ = OK, ✕ = Not OK		
6.19	Walls			
7.0	MEP schedules are created for:			
7.1	Multi-Category			
7.2	Assemblies			
8.0	BIM model modelled in accordance to BEP:			
8.1	Total number of models			
8.2	Model file names			
8.3	Models are linked correctly			
8.4	Grids and levels are correctly named.			
8.5	All survey and base point coordinates in the models are correct.			
8.6	Sheets from model table are correct.			
8.7	All element names are correct.			
8.8	All elements are using the correct category.			
8.9	All elements are modelled to the correct software format			
8.10	All elements are in the correct workset			
8.11	All MEP system types are setup correctly			
8.12	All MEP components are in their correct system types			
8.13	All system colors are correctly assigned.			
8.14	Parameters entered as listed.			

No.	Items	1st Check	Final Check	Reason/s
		✓ = OK, ✕ = Not OK		
9.0	Phase not used. (If yes, please state in detail.)			
10.0	No design options are used			
11.0	Unused items are purged			
12.0	Total file size			
13.0	Other formats delivered			
13.1	Individual models in .rvt, .nwc, .nwd, .ifc			
13.2	Federated model in .nwd			
13.3	All 2D drawings in .dwf			

Appendix - 3.2: BIM Model As-Built Acceptance Document Checklist**No.: A – 3.2**

Date : Year Month Day

Project No/Name: _____

First Check Date: _____ Micron Personnel Name: _____

Vendor Personnel Name: _____

Final check Date: _____ Micron Personnel Name: _____

Model: _____ Vendor Personnel Name: _____

No.	Items	1st Check	Final Check	Reason/s
		✓ = OK, ✕ = Not OK		
1.0	Provide laser scanning data mapped with model, or site photo for unable scan area. (If not available, please state reason/s.)			
2.0	Laser scan match 3D (If not match, please state reason/s.)			
3.0	The 2D drawings are direct produced from 3D.			
4.0	Maintenance area and clearance models included and acknowledged by FM O&M team. (Note down comments if any)			
5.0	Run Clash detection and show result. (Include clash matrix, graph, clash rules sets, clash report xml and clash detection model files for review.)			

6A	P&ID, SL, Schematics, Raiser, Systems diagrams match 3D model.			
6B	P&ID, SL, Schematics, Raiser, Systems diagrams tags are same in 3D model.			
Electrical				
6.1	List systems with P&ID, single line, schematic, flow, system, riser diagrams.			
Mechanical HVAC				
6.2	List systems with P&ID, single line, schematic, flow, system, riser diagrams.			
7.0	CSA schedules are created for:			
7.1	Multi-Category			
7.2	Assemblies			
7.3	Railings			
7.4	Railings Handrails			
7.5	Railings Support			
7.6	Railings Top Rails			
7.7	Ramps			
7.8	Roofs			
7.9	Roofs Fascias			
7.10	Roofs Gutters			
7.11	Roofs Soffits			
7.12	Site Pads			
7.14	Stair Landings			
7.15	Stair Runs			
7.16	Stair Supports			
7.17	Topography			
7.18	Walls Wall Sweeps			
7.19	Walls			

8.0	MEP schedules are created for:			
8.1	<i>Multi-Category</i>			
8.2	<i>Assemblies</i>			
9.0	BIM model modelled in accordance to BEP:			
9.1	Total number of models			
9.2	<i>Model file names</i>			
9.3	<i>Models are linked correctly</i>			
9.4	<i>Grids and levels are correctly named.</i>			
9.5	All survey and base point coordinates in the models are correct.			
9.6	<i>Sheets from model table are correct.</i>			
9.7	<i>All element names are correct.</i>			
9.8	<i>All elements are using the correct category.</i>			
9.9	<i>All elements are modelled to the correct software format</i>			
9.10	<i>All elements are in the correct workset</i>			
9.11	<i>All MEP system types are setup correctly</i>			
9.12	<i>All MEP components are in their correct system types</i>			
9.13	<i>All system colors are correctly assigned.</i>			
9.14	<i>Parameters entered as listed.</i>			
10.0	Phase not used. (If yes, please state in detail.)			
11.0	No design options are used			

12.0	Unused items are purged			
13.0	Total file size			
14.0				
14.1	Individual models in .rvt, .nwc, .nwd, .ifc			
14.2	Federated model in .nwd			
14.3	All 2D drawings in .dwf			

Appendix-4.0 Facility Process and Supply System EHS Checklist**No.: A – 4**

Date : Year Month Day

Purpose:

The purpose of this check list is to ensure that each completed system was being evaluated and confirmed on its compliance with EHS requirement prior to start-up. Facility should utilize this checklist to complete the inspection for EHS review and further confirmation.

System Name:**No.:****Location:**

Inspection of the Facility Process and Supply System EHS Checklist			
Name of system:		No.:	Location:
Item	Inspection method	Conformance/ nonconformance / NA	Improvement measures/ remarks
Supply of power Substation	1. The function of the local fire protection system has been confirmed and the test of the monitoring system has been completed.	☐ Yes ☐ No ☐ NA	
	2. The power supply panel has been labeled with warning sign provided in English and local language, e.g., Electrical Hazards, Arching, etc.	☐ Yes ☐ No ☐ NA	
	3. Tag Out/Lock Out mechanism was in-place.	☐ Yes ☐ No ☐ NA	
	4. Voltage rating has been labeled on the distribution box, terminal box and outlet.	☐ Yes ☐ No ☐ NA	
	5. Confirmation on air-conditioning and /or exhaust system for both indoor space and in the panels.	☐ Yes ☐ No ☐ NA	
	6. The interlock safety protection function had been tested.	☐ Yes ☐ No ☐ NA	
	7. Confirmation of design calculation and construction of seismic protection for electrical panel.	☐ Yes ☐ No ☐ NA	
	8. Illumination comply with all legislations and Micron requirement.	☐ Yes ☐ No ☐ NA	
	9. Access control and monitoring system for entrance to electrical room provided and in function.	☐ Yes ☐ No ☐ NA	

	10.Warning sign in English and local language provided at entrance to highlight hazards and warn off unauthorized entry.	☐ Yes ☐ No ☐ NA	
	11.PPE cabinet with arch flash clothing shall be provided in or nearby electrical room.	☐ Yes ☐ No ☐ NA	
	12.Emergency Exit clearly demarcated.	☐ Yes ☐ No ☐ NA	
Personnel/ date	Facility Host: Supervisor(M1):	EHS/ERC:	

Inspection of the Facility Process and Supply System EHS Checklist			
Name of system:		No.:	Location:
Item	Inspection method	Conformance/nonconformance / NA	Improvement measures/ remarks
Bulk Gas	1. Storage tank, cabinet, piping and support had been constructed as per seismic protection design calculation.	☐ Yes ☐ No ☐ NA	
	2. Confirmation on completion and readiness of all fireproofing coating and firestop.	☐ Yes ☐ No ☐ NA	
	3. Pipe/Tanks/Vessels labels (including the contents/direction of flow) have been provided according to Micron specification.	☐ Yes ☐ No ☐ NA	
	4. System which included storage tanks and piping have passed the required pressure test.	☐ Yes ☐ No ☐ NA	
	5. All piping openings must be sealed with blind flange (or cap) and labeled.	☐ Yes ☐ No ☐ NA	
	6. Testing of emergency cut-off mechanism (e.g., EMO) or relevant test reports.	☐ Yes ☐ No ☐ NA	
	7. Confirmation on function of overflow monitoring and alarm system.	☐ Yes ☐ No ☐ NA	
	8. Purification system safety interlock protection function had been tested.	☐ Yes ☐ No ☐ NA	
	9. Tanks/Vessels/Vaults labels (including the contents/direction of flow) have been provided on confined space.	☐ Yes ☐ No ☐ NA	
Personnel/ date	Facility Host: Supervisor(M1):	EHS/ERC:	

Inspection of the Facility Process and Supply System EHS Checklist			
Name of system:		No.:	Location:
Item	Inspection method	Conformance/ Nonconformance / NA	Improvement measures/ remarks
Specialty Gas Supply	1. Storage tank, cabinet, piping and support had been constructed as per seismic protection design calculation.	☐ Yes ☐ No ☐ NA	
	2. Confirmation on completion and readiness of all fireproofing coating and firestop.	☐ Yes ☐ No ☐ NA	
	3. Confirm all joints on piping are properly connected and leak test completed.	☐ Yes ☐ No ☐ NA	
	4. Pipe/Tanks/Vessels labels (including the contents/direction of flow) have been provided according to Micron specifications.	☐ Yes ☐ No ☐ NA	
	5. Every valve has a ON/OFF tag.	☐ Yes ☐ No ☐ NA	
	6. Coding/labeling on piping at supply end and terminal end must be the same.	☐ Yes ☐ No ☐ NA	
	7. All piping openings must be sealed with blind flange (or cap) and label.	☐ Yes ☐ No ☐ NA	
	8. All joints and valves shall be located in VMB which is equipped with exhaust (photohelic / magnehelic monitoring).	☐ Yes ☐ No ☐ NA	
	9. Emergency scrubber function test conducted.	☐ Yes ☐ No ☐ NA	
	10. Emergency exhaust system function tested.	☐ Yes ☐ No ☐ NA	
	11. Automatic and manual cut-off system (EMO) function tested.	☐ Yes ☐ No ☐ NA	
	12. Both local and on-line gas detectors function tested.	☐ Yes ☐ No ☐ NA	
	13. Confirmation on design of sprinkle system water volume for combustible gas supply and function test of UV/IR interlocking system.	☐ Yes ☐ No ☐ NA	

	14.Indoor or in-equipment/in-tool fire system tested.	☐ Yes ☐ No ☐ NA	
	15.Confirmation on the provision and specifications of the explosion relief panels.	☐ Yes ☐ No ☐ NA	
	16.The indoor or in-equipment/in-tool fire protection system has been completed (no sprinkling system is available in the area whereby water is prohibited).	☐ Yes ☐ No ☐ NA	
	17.Cylinder emergency cut-off device.	☐ Yes ☐ No ☐ NA	
	18.The interlock safety protection function tested.	☐ Yes ☐ No ☐ NA	
	19.Access control system working.	☐ Yes ☐ No ☐ NA	
	20.Emergency Exit clearly demarcated.	☐ Yes ☐ No ☐ NA	
	21.LOTO capability for valves.	☐ Yes ☐ No ☐ NA	
	22.Emergency Eyewash and Shower function test.	☐ Yes ☐ No ☐ NA	
Personnel / date	Facility Host:	Supervisor(M1):	EHS/ERC:

Inspection of the Facility Process and Supply System EHS Checklist			
Name of system:		No.:	Location:
Item	Inspection method	Conformance/ Nonconformance/ NA	Improvement measures/ remarks
Chemical Supply	1. Storage tank, CDS, piping and support had been constructed as per seismic protection design calculation.	Y ☐ N ☐ NA ☐	
	2. Confirmation on completion and readiness of all fireproofing coating and firestop.	Y ☐ N ☐ NA ☐	
	3. Confirm all joints on piping are properly connected, and leak test completed.	Y ☐ N ☐ NA ☐	
	4. Pipe/Tanks/Vessels labels (including the contents/direction of flow) have been provided according to Micron specifications.	Y ☐ N ☐ NA ☐	
	5. Every valve has a ON/OFF tag.	Y ☐ N ☐ NA ☐	
	6. Coding/labeling on piping at supply end and terminal end must be the same.	Y ☐ N ☐ NA ☐	
	7. Both local, and on-line liquid sensor function tested.	Y ☐ N ☐ NA ☐	
	8. All piping openings must be sealed with blind flange (or cap) and label.	Y ☐ N ☐ NA ☐	
	9. Piping for chemicals shall be double lining piping(inorganic) or SUS316 (organic).	Y ☐ N ☐ NA ☐	
	10.All joints and valves shall be located in VMB which is equipped with exhaust (photohelic / magnehelic monitoring).	Y ☐ N ☐ NA ☐	
	11.Confirmation on design and function test of Fire suppression system use on combustible chemical.	Y ☐ N ☐ NA ☐	
	12.Confirmation on capacity of the overflow containment and completion of the emergency discharge system.	Y ☐ N ☐ NA ☐	
	13.Materials used for supply system of combustible contents shall be compatible.	Y ☐ N ☐ NA ☐	

	14.Materials used for supply system on acid/alkali shall be of anti-corrosion materials.	Y ☐ N ☐ NA ☐	
	15.Confirmation on the exhaust volume of the indoor air conditioner, and the height of the extractor opening.	Y ☐ N ☐ NA ☐	
	16.Functional test for interlock safety protection completed.	Y ☐ N ☐ NA ☐	
	17.Secondary containment for locations where chemical delivery takes place.	Y ☐ N ☐ NA ☐	
	18.Emergency Eyewash and Shower function test.	Y ☐ N ☐ NA ☐	
	19.LOTO capability for valves.	Y ☐ N ☐ NA ☐	
	20.Tanks/Vessels/Vaults labels (including the contents/direction of flow) have been provided on confined space.	Y ☐ N ☐ NA ☐	
	21.Emergency Exit clearly demarcated.	Y ☐ N ☐ NA ☐	
	22.Emergency Eyewash and Shower function test.	Y ☐ N ☐ NA ☐	
Personnel/ date	Facility Host: Supervisor(M1):	EHS/ERC:	

Inspection of the Facility Process and Supply System EHS Checklist			
Name of system:		No.:	Location:
Item	Inspection method	Conformance/ nonconformance/ NA	Improvement measures/ remarks
Water (PCW/ JW/ PW/ UPW/ HPW, etc.)	1. Storage tank, piping and support had been constructed as per seismic protection design calculation.	Y ☐ N ☐ NA ☐	
	2. Confirmation on completion and readiness of all fireproofing coating and firestop.	Y ☐ N ☐ NA ☐	
	3. Confirm that all pipe joints are properly connected, and leak test completed.	Y ☐ N ☐ NA ☐	

	4. Pipe labels (including the contents/direction of flow) have been provided according to Micron specifications.	Y ☐ N ☐ NA ☐	
	5. All piping openings must be sealed with blind flange (or cap) and label.	Y ☐ N ☐ NA ☐	
	6. Ensure water pipes are not installed above electrical panels, and all measures are done to avoid water pipe above electrical panels.	Y ☐ N ☐ NA ☐	
	7. LOTO capability for valves.	Y ☐ N ☐ NA ☐	
	8. Tanks/Vessels/Vaults labels (including the contents/direction of flow) have been provided on confined space.	Y ☐ N ☐ NA ☐	
Personnel/ date	Facility Host:	Supervisor(M1):	EHS/ERC:

Inspection of the Facility Process and Supply System EHS Checklist			
Name of system:		No.:	Location:
Item	Inspection method	Conformance/ Nonconformance / NA	Improvement measures/ remarks
Drain (wastewater, waste acid, waste alkali, waste solvent, etc.)	1. Storage tank, piping and support had been constructed as per seismic protection design calculation.	Y ☐ N ☐ NA ☐	
	2. Confirmation on completion and readiness of all fireproofing coating and firestop.	Y ☐ N ☐ NA ☐	
	3. Confirm that all pipe joints are properly connected, and leak test completed.	Y ☐ N ☐ NA ☐	
	4. Pipe labels (including the contents/direction of flow) have been provided according to Micron specifications.	Y ☐ N ☐ NA ☐	
	5. All piping openings must be sealed with blind flange (or cap) and label.	Y ☐ N ☐ NA ☐	
	6. Confirmation on completion of installation and leak test for the collection tank.	Y ☐ N ☐ NA ☐	

	7. Function test of the local exhaust system has been completed.	Y ☐ N ☐ NA ☐	
	8. The localize fire protection system has been completed.	Y ☐ N ☐ NA ☐	
	9. Function test of the leak preventive devices and emergency exhaust installation has been completed.	Y ☐ N ☐ NA ☐	
	10. The interlock safety protection function tested.	Y ☐ N ☐ NA ☐	
	11. Ensure water pipes are not installed above electrical panels.	Y ☐ N ☐ NA ☐	
	12. LOTO capability for valves.	Y ☐ N ☐ NA ☐	
	13. Emergency Eyewash and Shower function test.	Y ☐ N ☐ NA ☐	
Personnel/ date	Facility Host: Supervisor(M1):	EHS/ERC:	

Inspection of the Facility Process and Supply System EHS Checklist			
Name of system:		No.:	Location:
Item	Inspection method	Conformance/ Nonconformance / NA	Improvement measures/ remarks
Central scrubber	1. Storage tank, piping and support had been constructed as per seismic protection design calculation.	Y ☐ N ☐ NA ☐	
	2. Confirmation on completion and readiness of all fireproofing coating and firestop.	Y ☐ N ☐ NA ☐	
	3. Confirm that all pipe joints are properly connected, and leak test completed.		
	4. Pipe labels (including the contents/direction of flow) have been provided according to Micron specifications.	Y ☐ N ☐ NA ☐	
	5. Functional test of the damper completed.	Y ☐ N ☐ NA ☐	
	6. Installation of the sampling port has been completed.	Y ☐ N ☐ NA ☐	
	7. All piping openings must be sealed with blind flange (or cap) and label.	Y ☐ N ☐ NA ☐	
	8. The scrubber must be made from non-combustible material or the sprinkling function at the inlet has been provided and tested.	Y ☐ N ☐ NA ☐	

	9. Function test of the exhaust and chemical dosing system has been completed.	Y ☐ N ☐ NA ☐	
	10. The function test of the scrubber overflow containment and emergency exhaust installation has been completed.	Y ☐ N ☐ NA ☐	
	11. Function test of drain port completed.	Y ☐ N ☐ NA ☐	
	12. The interlock safety protection function tested.	Y ☐ N ☐ NA ☐	
	13. LOTO capability for valves.	Y ☐ N ☐ NA ☐	
	14. Tanks/Vessels/Vaults labels (including the contents/direction of flow) have been provided on confined space.	Y ☐ N ☐ NA ☐	
	15. Emergency Exit clearly demarcated.	Y ☐ N ☐ NA ☐	
	16. Emergency Eyewash and Shower function test.	Y ☐ N ☐ NA ☐	
Personnel/ date	Facility Host: Supervisor(M1):	EHS/ERC:	

Inspection of the Facility Process and Supply System EHS Checklist			
Name of system:		No.:	Location:
Item	Inspection method	Conformance/ Nonconformance / NA	Improvement measures/ remarks
VOC treatment system	1. Storage tank, piping and support had been constructed as per seismic protection design calculation.	Y ☐ N ☐ NA ☐	
	2. Confirmation on completion and readiness of all fireproofing coating and firestop.	Y ☐ N ☐ NA ☐	
	3. Confirm that all pipe joints are properly connected, and leak test completed.	Y ☐ N ☐ NA ☐	
	4. Pipe labels (including the contents/direction of flow) have been provided according to Micron specifications.	Y ☐ N ☐ NA ☐	
	5. Function test for the emergency cut-off of combustion system has been completed.	Y ☐ N ☐ NA ☐	
	6. Function test of the localize exhaust system has been completed.	Y ☐ N ☐ NA ☐	
	7. Both internal and localize fire protection systems have been completed.	Y ☐ N ☐ NA ☐	
	8. Function test of the leak detectors and emergency exhaust facilities has been completed.	Y ☐ N ☐ NA ☐	
	9. Function test on performance and monitoring system completed.	Y ☐ N ☐ NA ☐	
	10. Function test for scrubber backup and on-line system function completed.	Y ☐ N ☐ NA ☐	
	11. Function test for the monitor and emergency cut-off of the fuel supply system completed.	Y ☐ N ☐ NA ☐	
	12. The interlock safety protection function tested.	Y ☐ N ☐ NA ☐	
	13. LOTO capability for valves.	Y ☐ N ☐ NA ☐	
	14. Tanks/Vessels/Vaults labels (including the contents/direction of flow) have been provided on confined space.	Y ☐ N ☐ NA ☐	
Personnel/ date	Facility Host: Supervisor(M1):	EHS/ERC:	

Inspection of the Facility Process and Supply System EHS Checklist			
Name of system:		No.:	Location:
Item	Inspection method	Conformance/ Nonconformance/ NA	Improvement measures/ remarks
Natural Gas Supply	1. Storage tank, piping and support had been constructed as per seismic protection design calculation.	Y ☐ N ☐ NA ☐	
	2. Confirmation on completion and readiness of all fireproofing coating and firestop.	Y ☐ N ☐ NA ☐	
	3. Confirm that all pipe joints are properly connected, and leak test completed.	Y ☐ N ☐ NA ☐	
	4. Pipe labels (including the contents/direction of flow) have been provided according to Micron specifications.	Y ☐ N ☐ NA ☐	
	5. Every valve in the pipe and supply cabinet has a tag indicating ON or OFF.	Y ☐ N ☐ NA ☐	
	6. Every supplying pipe of the cabinet has been labeled with the code of the corresponding tool.	Y ☐ N ☐ NA ☐	
	7. All piping openings must be sealed with blind flange (or cap) and label.	Y ☐ N ☐ NA ☐	
	8. Function test for both automatic and manual cut-off mechanism of the valve manifold box completed.	Y ☐ N ☐ NA ☐	
	9. Function test for gas detector installed at gas pipes passing through a confined space completed.	Y ☐ N ☐ NA ☐	
	10. The interlock safety protection function tested.	Y ☐ N ☐ NA ☐	
	11. LOTO capability for valves.	Y ☐ N ☐ NA ☐	
Personnel/ date	Facility Host: Supervisor(M1):	EHS/ERC:	

Conclusion:						
EHS Sponsor	Site Project Construction Direct Manager	Site Project Operation Direct Manager	Site Project Construction System Sponsor	Site Project Operation System Sponsor	GCT Engineering Management	Contractor

Appendix-5.0 Design in Safety Checklist and Maintainability Guide

This checklist has been developed by applying the hierarchy of control – in order of priority:

- 1. Eliminate** – design the hazard out of the building or structure.
- 2. Substitute** – less hazardous materials, fixtures, fittings, plant or construction methods.
- 3. Isolate** – use guards or barriers to limit access to hazard.
- 4. Engineering** – minimize risk by engineering means, eg provide a permanent building maintenance.
- 5. Administrative controls** – recommend the establishment of systems of work or signage, where required, to control residual risks.
- 6. Personal protective equipment** – recommend suitable personal protective equipment and training, where required, to control residual risks.

A combination of these measures has been applied when no single measure is enough to eliminate or minimize the risk.

The checklist has been completed incorporating consultation from relevant stakeholders, such as the clients and project managers.

This checklist assesses risks for only those components and elements included in the original design. Persons using this report must determine if additional elements have been added to the project that may impact on, alter, or be additional to, the hazards identified in this checklist. While due care has been taken to review hazards in relation to the project during the design process it should be reviewed prior to any construction works commencing and updated as required.

Where actions and comments are recorded in the far-right column, these points are not written with the intent of Directing persons giving effect to the design (construction phase) or those with the responsibility for maintenance operation or disposal of the asset after practical completion, but rather to provide recommendations for consideration that may assist in reduction of risks where hazards or impacts have been identified by the designer.

No.: A – 5.1 Design in Safety Checklist

Date : Year Month Day

Project Name:	<i>Insert Project Number and Title</i>
Prepared by:	<i>Insert Name and Position Title</i>
Date:	#

Items for consideration (how will it be built in a safe way, how will it be used in a safe way)	Potential design issues during construction	Potential design issues during operation	Comments / actions specific to the works occurring
Site Layout			
Have overhead power/underground services been located? <i>Eg. Dial before you dig.</i>	Yes ☐ No ☐	Yes ☐ No ☐ N/A ☐	
Will materials and equipment be stored in a convenient location?	Yes ☐ No ☐	Yes ☐ No ☐ N/A ☐	
Are there any other specific site layout issues that need to be considered?	Yes ☐ No ☐	Yes ☐ No ☐ N/A ☐	

Items for consideration (how will it be built in a safe way, how will it be used in a safe way)	Potential design issues during construction	Potential design issues during operation	Comments / actions specific to the works occurring
Heights			
Can anyone be injured by falling a distance of 1.8 meters from either an elevation or into an excavation Consider raised road surfaces and trenching / excavations during construction and after practical completion.	Yes ☐ No ☐	Yes ☐ No ☐ N/A ☐	The contractor to ensure that temporary fall prevention measures have been installed while works are being undertaken, prior to permanent fall protection devices (ie. Guard rails) are installed.
Can anyone be injured by objects falling from a height? <i>Eg. Work platforms, ladders.</i>	Yes ☐ No ☐	Yes ☐ No ☐ N/A ☐	
Timing			
Has the effect of the environment been considered when planning the sequence of construction for long term jobs? <i>Eg. Wet/Dry season considerations</i>	Yes ☐ No ☐	Yes ☐ No ☐ N/A ☐	
Access/Egress			
Will vehicles / mobile plant have easy access to the site?	Yes ☐ No ☐	Yes ☐ No ☐ N/A ☐	

Items for consideration (how will it be built in a safe way, how will it be used in a safe way)	Potential design issues during construction	Potential design issues during operation	Comments / actions specific to the works occurring
Environmental Conditions			
What flora and fauna need to be considered in the local environment?	Yes ☐ No ☐	Yes ☐ No ☐ N/A ☐	
Are there approvals and sufficient resources for gravel and water extraction?	Yes ☐ No ☐	Yes ☐ No ☐ N/A ☐	
What sediment and erosion control issues exist in the local environment? <i>i.e. table drains, run off, waterways, drainage?</i>	Yes ☐ No ☐	Yes ☐ No ☐ N/A ☐	
Will dust suppression be an issue during the works and after practical completion?	Yes ☐ No ☐	Yes ☐ No ☐ N/A ☐	
Will the noise during construction significantly disrupt surrounding residents or businesses? <i>Consider nearby sensitive receptors such as schools, hospitals etc.</i>	Yes ☐ No ☐	Yes ☐ No ☐ N/A ☐	

Items for consideration (how will it be built in a safe way, how will it be used in a safe way)	Potential design issues during construction	Potential design issues during operation	Comments / actions specific to the works occurring
Lighting to be installed for night time work?	Yes ☐ No ☐	Yes ☐ No ☐ N/A ☐	
Environmental Approvals			
Have areas of cultural significance been considered?	Yes ☐ No ☐	Yes ☐ No ☐ N/A ☐	
Are there land tenure and land council issues to be addressed?	Yes ☐ No ☐	Yes ☐ No ☐ N/A ☐	
Are there any conditions in the environment approval that needs to be considered?	Yes ☐ No ☐	Yes ☐ No ☐ N/A ☐	
Public Security / Traffic Control			
Will access to the site by members of the public be restricted to prevent injury and vandalism? <i>Eg barricades or fences, locked access ways</i>	Yes ☐ No ☐	Yes ☐ No ☐ N/A ☐	Contractor to ensure that all plant and equipment is stored in a secure location during site works.
Will traffic need to be controlled around the site?	Yes ☐ No ☐	Yes ☐ No ☐ N/A ☐	Contract to develop a Traffic Management Plan an approved prior to any site works commencing.

Items for consideration (how will it be built in a safe way, how will it be used in a safe way)	Potential design issues during construction	Potential design issues during operation	Comments / actions specific to the works occurring
Will loading or unloading of materials and equipment be needed on site?	Yes ☐ No ☐	Yes ☐ No ☐ N/A ☐	
Traffic control sufficient to provide access emergency vehicles? <i>I.e. Fire, police and ambulance?</i>	Yes ☐ No ☐	Yes ☐ No ☐ N/A ☐	
HAZCHEM			
Could members of the public be exposed to hazardous material, dust, vapours etc?	Yes ☐ No ☐	Yes ☐ No ☐ N/A ☐	
Could anyone be injured due to exposure to asbestos, lead or other hazardous materials?	Yes ☐ No ☐	Yes ☐ No ☐ N/A ☐	
Will hazardous/flammable substances/chemicals be stored correctly? <i>Eg. appropriate type and size of containers</i>	Yes ☐ No ☐	Yes ☐ No ☐ N/A ☐	Ensure all chemicals are handled in accordance with the Safety Data Sheet. Storage containers are to be of appropriate type and size for substances they hold and the use they have on site.

Items for consideration (how will it be built in a safe way, how will it be used in a safe way)	Potential design issues during construction	Potential design issues during operation	Comments / actions specific to the works occurring
Will there be adequate ventilation available?	Yes ☐ No ☐	Yes ☐ No ☐ N/A ☐	
Fire / Explosion			
Will surfaces finishes be fire resistant? Material used, can they be destroyed by fire?	N/A	Yes ☐ No ☐ N/A ☐	
Working Environment			
Is there sufficient ventilation to meet requirements for the work to be performed on the project site?	Yes ☐ No ☐	Yes ☐ No ☐ N/A ☐	
What will be the weather conditions on site? <i>ie: extreme hot or cold weather, dusty or windy conditions, storms.</i>	Yes ☐ No ☐	Yes ☐ No ☐ N/A ☐	The contractor is responsible for developing a safe work method for working in site weather conditions to provide safe work conditions.
Is there potential for the site to flood or be subject to inundation of water?	Yes ☐ No ☐	Yes ☐ No ☐ N/A ☐	
Is there a risk to workers from flora or fauna?	Yes ☐ No ☐	Yes ☐ No ☐ N/A ☐	

Items for consideration (how will it be built in a safe way, how will it be used in a safe way)	Potential design issues during construction	Potential design issues during operation	Comments / actions specific to the works occurring
Will the workers have access to amenities and facilities such as storage, first aid rooms, rest rooms, meal and accommodation areas and drinking water?	Yes ☐ No ☐	Yes ☐ No ☐ N/A ☐	The contractor is to ensure that adequate facilities are provided.
Will workers be exposed to noise from plant or aspects at the project site?	Yes ☐ No ☐	Yes ☐ No ☐ N/A ☐	
Utilities and Services			
Will the location to, access to and egress from facilities, site office or other be an issue on the project?	Yes ☐ No ☐	Yes ☐ No ☐ N/A ☐	
Building Materials / Structural Safety			
Will the structural strength and stability be enough to hold permanent and temporary loads?	Yes ☐ No ☐	Yes ☐ No ☐ N/A ☐	
Will there be procedures for disposal of waste?	Yes ☐ No ☐	Yes ☐ No ☐ N/A ☐	Procedures are to be developed for the temporary on site storage of materials, and the transportation and disposal methods.

Items for consideration (how will it be built in a safe way, how will it be used in a safe way)	Potential design issues during construction	Potential design issues during operation	Comments / actions specific to the works occurring
Confined Spaces			
Will anyone need to work in a confined space?	Yes ☐ No ☐	Yes ☐ No ☐ N/A ☐	
Will adequate ventilation be provided in confined spaces?	Yes ☐ No ☐	Yes ☐ No ☐ N/A ☐	
Emergency Procedures			
Will evacuation and emergency procedures be developed and clearly communicated?	Yes ☐ No ☐	Yes ☐ No ☐ N/A ☐	The successful contractor to develop a site specific Safety Management Plan that deals with fires during construction. Location, size and type of firefighting equipment on site to commensurate with potential fire risks at the site. Firefighting equipment to be compliant and serviced. Local fire station UHF channel and phone numbers for emergency services to be obtained and held at the site office. Site and responsible person details to be provided to the closest emergency service.
Site Access			

Items for consideration (how will it be built in a safe way, how will it be used in a safe way)	Potential design issues during construction	Potential design issues during operation	Comments / actions specific to the works occurring
Will site lines provide adequate visibility for trucks/cars entering/exiting the site?	Yes ☐ No ☐	Yes ☐ No ☐ N/A ☐	
Will the access road have adequate stability for trucks/equipment?	Yes ☐ No ☐	Yes ☐ No ☐ N/A ☐	
Has adequate turning room been provided for large vehicles and trucks?	Yes ☐ No ☐	Yes ☐ No ☐ N/A ☐	
Will the strength and stability of the road or bridge structure be enough to hold permanent and temporary loads?	Yes ☐ No ☐	Yes ☐ No ☐ N/A ☐	
Excavation			
Has the stability of walls been checked and support provided where required?	Yes ☐ No ☐	Yes ☐ No ☐ N/A ☐	
Has a method of access/egress been provided at spillway stilling basin for maintenance?	N/A	Yes ☐ No ☐ N/A ☐	

Items for consideration (how will it be built in a safe way, how will it be used in a safe way)	Potential design issues during construction	Potential design issues during operation	Comments / actions specific to the works occurring
Dams and Waterways			
Has the stability of walls been checked and support provided where required?	Yes ☐ No ☐	Yes ☐ No ☐ N/A ☐	
Is there a safe manual access point for sample collection?	N/A	Yes ☐ No ☐ N/A ☐	
Will adequate covers and railings be provided to prevent falls?	Yes ☐ No ☐	Yes ☐ No ☐ N/A ☐	
Pipelines			
Will equipment such as pipes be securely stored to prevent rolling, theft etc?	Yes ☐ No ☐	Yes ☐ No ☐ N/A ☐	
Was the location of the pipeline selected to minimise disruption to congested or public areas during construction and/or maintenance?	Yes ☐ No ☐	Yes ☐ No ☐ N/A ☐	
Has the distance/proximity to other existing services be considered during the selection of the location of the pipeline?	Yes ☐ No ☐	Yes ☐ No ☐ N/A ☐	
Are supports required in excavations?	Yes ☐ No ☐	Yes ☐ No ☐ N/A ☐	

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Items for consideration (how will it be built in a safe way, how will it be used in a safe way)	Potential design issues during construction	Potential design issues during operation	Comments / actions specific to the works occurring
If old pipelines are being replaced, will old pipelines be removed or remain capped in the ground?	Yes ☐ No ☐	Yes ☐ No ☐ N/A ☐	

Other Issues			

What are the next steps when the Checklist has been completed?

1. Where hazards are identified, make changes to the design whenever reasonably practicable and record in comments and actions column. In identifying changes, apply the hierarchy of control so that the risk is eliminated or, if not reasonably practicable to do so, minimised to the lowest level reasonably practicable.
2. Provide a completed copy of this review checklist to the Project Manager. This information will be included in the contract documentation so that the hazards are brought to the attention of tenderers/ the contractor and addressed in their Safety Management Plan.

Name of
Reviewer_____
Position held by reviewer_____
Signature of Reviewer_____
Date of Review

No.: A – 5.2 Maintainability Guide

Date : Year Month

Day

Use Instruction The designer should indicate “Y”, “N” or “NA” in the Y/N/NA column against the design recommendations.

Abbreviation	Denotes
Y	Yes – meet or exceed the design recommendation
N	No – does not meet the design recommendation
NA	Not Applicable – design recommendation is not applicable

The designer should provide a brief description of the provision. If the design recommendation is not met or not applicable, the designer should explain why the recommendation is not considered and whether alternative solution is proposed. Additional notes, references and drawings could be attached to this guide as supporting documents.

As the design objectives differ from one building to another, the recommendations of the guide may not comprehensively address the scope and specific maintenance requirements of a project. The designer should use this guide in conjunction with other applicable codes, regulations, and design guidelines.

An example of completed checklist is as follows:

A1.	Access	Y / N / NA	Description of provision (Attach relevant drawings or references, etc. where necessary)	If No or Not Applicable, please explain
A1.1	Protruding façade features Avoid extensive niches, fins and ledges that protrude more than 600mm. If the protrusions exceed 600mm, designers should make specific considerations for safe and easy access.	NA		No protruding features on façade.
A1.2	Internal Access Façade design should promote minor cleaning and repair works to be carried out from within the building, while major repair works can take place from the outside. Use modularised window panels which are not too large (max 750mm) or reversible windows for ease of cleaning from within the building, i.e. within reach of a cleaner's arm and his/her handheld tools.	Y	Gondola system and elevated walkway access provided	
A1.3	Access to any part of the façade should not encroach on private spaces or require deconstruction of window units or other building components.	Y	No encroachment into private spaces	
A1.4	Building Maintenance Unit (BMU) Where BMU is used for façade maintenance, ensure that the BMU is: a) electronically powered and programmable.	N		No BMU system is used. See 'façade access strategy' described in the Maintenance Strategy Report

1. Architecture
- 1.1. Access to facades

Objectives:

Provide safe and easy access to every part of a building's façade and all elements therein (e.g., sunshades, canopies, claddings, lights, signs, and other façade features) – for efficient cleaning and maintenance, and inspection.

1.	<u>Access</u>	Y / N / NA	Description of provision <i>(Attach relevant drawings, references, etc. where necessary)</i>	If No or Not Applicable, please explain
1.1	Protruding façade features Avoid extensive niches, fins and ledges that protrude more than 600mm. If the protrusions exceed 600mm, designers should make specific considerations for safe and easy access.			
1.2	Internal Access Façade design should promote minor cleaning and repair works to be carried out from within the building, while major repair works can take place from the outside. Use modularized window panels which are not too large (max 750mm) or reversible windows for ease of cleaning from within the building, i.e., within reach of a cleaner's arm and his/her handheld tools.			
1.3	Access to any part of the façade should not encroach on private spaces or require deconstruction of window units or other building components.			
1.4	Building Maintenance Unit (BMU) Where BMU is used for façade maintenance, ensure that the BMU is: a) Electronically powered and programmable. b) Designed to enable positioning/suspension of working platforms to reach all exterior surfaces of the building. c) Easily accessible via common areas. d) Designed with suitable restraints and bumpers to address wind loads.			

1.	<u>Access</u>	Y / N / NA	Description of provision <i>(Attach relevant drawings, references, etc. where necessary)</i>	If No or Not Applicable, please explain
1.5	Suspended Working Platform Where suspended working platform (gondola) is used for façade maintenance, ensure that: a) There are provisions for gondola to be safely affixed to the building (e.g., permanent installation points) as well as adequate anchor points for safety ropes and independent anchor for lifelines. b) The gondola is able to reach all parts of the building facade. c) Façade features do not obstruct the operation of the gondolas. Surfaces below protrusions/ cantilevered structures should be reachable and maintainable from the gondola. d) There are adequate safe launching/ landing places for the cradle/ working platform. e) The launching and landing spaces and the access routes to these spaces are located within common spaces and should avoid landing in landscaped areas. f) There is provision of power supply for gondola operations and water supply for carrying out maintenance works. g) Ensure spacing between external walls/ structures of adjacent blocks is at least 1.2m to facilitate the safe use of gondolas.			
1.6	Ground-based (Mobile Elevating Work Platform (MEWP)) Where MEWP is used for façade maintenance, ensure that: a) Unobstructed access route and working space are provided, b) Access route and working space should be level and designed to take the equipment's load.			
1.7	Provision for use of ladders Provide firm and level surface for use of ladders.			

1.	<u>Access</u>	Y / N / NA	Description of provision <i>(Attach relevant drawings, references, etc. where necessary)</i>	If No or Not Applicable, please explain
1.8	Safety of Maintenance Personnel Provide safe working environment for the setting up and operation of access systems such as: a) Railing, b) Anchorage points and/ or lifelines for sliding safety harness, c) Eyebolt fixing for rope to secure ladder.			
1.9	Air Conditioning (AC) Unit Service Ledge Provide sufficient working space and safe working environment (e.g., anchorage for safety harness) for maintenance personnel.			
1.10	Where maintenance access to the service ledge is via windows or wall apertures, the size of the openings should allow safe and easy access for maintenance personnel carrying tools, equipment, and component parts. Maintenance activities should not require the dismantling of services and building elements.			
1.11	Provide safety barriers or guard rails (minimum height of 1m) around the service ledge.			

1.2. Choice of Materials & Finishes for Facades

Objectives:

Use appropriate materials and finishes that are resistant to deterioration and exposure to weather elements.

2.	Materials & Finishes	Y / N / NA	Description of provision (Attach relevant drawings, references, etc. where necessary)	If No or not applicable, please explain
2.1	Adopt materials and finishes that are readily available (i.e., does not require special procurement).			
2.2	Paint Systems Use paint systems that enhance weather protection and lengthen the cyclical period needed for façade re- painting (e.g., paint with anti-stain and self-cleansing properties).			
2.3	Fastenings and Fixings Use materials that are not prone to corrosion or rust e.g., stainless steel.			
2.4	Outdoor Materials Avoid materials that are prone to deterioration when exposed to weathering (e.g., plaster or calcium silicate board).			
2.5	Sealants Use sealants that are resistance to staining and bleeding.			

1.3. Design and detailing for facades

Objectives:

Provide effective design and detailing to promote efficient run-off on façade surfaces to minimize water ponding, penetration, and staining problems.

3.	<u>Design & Detailing</u>	Y / N / NA	Description of provision (Attach relevant drawings, references, etc. where necessary)	If No or not applicable, please explain
3.1	External Walls Provide drip edges to prevent streaks on exterior soffits, walls, and glazed surfaces.			
3.2	Grade all top surfaces of walls to fall away from external face of walls to minimize streaks on the facade. The coping should overhang the rear side of the walls and comes with drip control to mitigate streaks on the back walls.			
3.3	Stone Panels Stone panels should be mechanically fixed. Avoid the use of adhesive which are prone to efflorescence.			
3.4	Install stone panels in a way that allows for inspection of its concealed mechanical connections to pre-empt corrosion or defects (e.g., accessible for fiber-optics inspection).			
3.5	Metal Cladding and Fixtures Avoid metal cladding and wrapping around external building components where high pedestrian traffic is expected.			
3.6	Install metal cladding in a way that allows for inspection of its concealed mechanical connections to pre-empt corrosion, dilapidation and loose or defective fixings.			
3.7	Glazed Surfaces Avoid inclined glass surfaces, being prone to dirt and dust collection.			
3.8	Size up external glazing optimally such that glass panels can be transported by lifts to facilitate future replacement from within building.			

3.	<u>Design & Detailing</u>	Y / N / NA	Description of provision (Attach relevant drawings, references, etc. where necessary)	If No or not applicable, please explain
3.9	Avoid the use of non-planar or curvilinear glass surfaces which are difficult to maintain or not readily available in the market.			
3.10	Ensure that glass (e.g., glazed canopies) are able to withstand the loads imposed during maintenance.			
3.11	Glass Parapets Avoid extensive use of glass parapet walls at outdoor areas as weather stains on glass surfaces are more visible and require more frequent cleaning.			
3.12	Weather Control Devices Incorporate appropriate weather control devices (such as overhanging ledges, copings, and rain screens) to external openings to minimize rainwater ingress. Where wall openings/doors are subject to external weathering, incorporate details such as overhang at the top, and water-bar or threshold kerb at the bottom to minimize water ingress.			
3.13	Joints Provide movement joints to accommodate movement in large continuous areas, or between adjacent building components (e.g., brick wall and concrete column) and dissimilar substrates. The backer-rod material in the joint should be compatible with the sealant used.			
3.14	Anti-Roosting Measures Adopt designs that mitigate roosting of birds (e.g., use chamfered or steeply sloped surfaces).			
3.15	Planter boxes Planter boxes should be accessible for easy maintenance (e.g., located at windowsill level, along service balcony/ corridor).			

3.	<u>Design & Detailing</u>	Y / N / NA	Description of provision (Attach relevant drawings, references, etc. where necessary)	If No or not applicable, please explain
3.16	The location of drainage outlets of planters should not cause stains on building facade.			
3.17	Loading/ Unloading Areas Provide protection to beams and columns in loading/ unloading areas (e.g., rubberized sleeves).			

1.4. Roof Access

Objectives:

Design the roof space for safe and efficient movement of maintenance personnel carrying tools, equipment, and spare parts.

4.	<u>Access</u>	Y / N / NA	Description of provision (Attach relevant drawings, references, etc. where necessary)	If No or not applicable, please explain
4.1	Direct Access Provide at least one direct staircase access to main roof areas for maintenance purpose.			
4.2	Safety Provide safety barriers/guard rails (min height of 1.0m) or anchor points for safety harness along the roof perimeter and roof openings (such as skylights and air wells) to prevent fall from height.			
4.3	Provide minimum 1.5m clear working space (or larger as specified by M&E equipment supplier) between the building edge/parapet wall and M&E installations.			
4.4	Provide non-slip steps/platforms/catwalks over/along maintenance path to prevent stepping on rooftop services such as piping/ trunking/ conduit.			
4.5	Provide clear demarcation of maintenance boundaries and access at roof areas.			
4.6	Provide hoisting facilities and anchorages at the roof for hoisting of			

4.	<u>Access</u>	Y / N / NA	Description of provision (Attach relevant drawings, references, etc. where necessary)	If No or not applicable, please explain
	equipment after completion of the building.			
4.7	Where façade extends more than 5m above the roof, provide catwalk or other means of access at the internal side of the façade.			

1.5. Materials and finishes for roofs

Objectives:

Use appropriate materials and finishes that are resistant to deterioration and exposure to weather elements.

5.	<u>Materials and Finishes</u>	Y / N / NA	Description of provision (Attach relevant drawings, references, etc. where necessary)	If No or not applicable, please explain
5.1	Provide adequate waterproofing to roof surfaces			
5.2	Avoid extensive use of transparent and translucent roofing materials that are prone to discoloration and deterioration when exposed to weather elements.			
5.3	Use roofing materials/finishes that are able to withstand heavy maintenance works (e.g., removing of algae).			

1.6. Roof design and detailing

Objectives:

Provide effective design and detailing to ensure optimal run-off to avoid water ponding and penetration problems.

6.	<u>Design & Detailing</u>	Y / N / NA	Description of provision (Attach relevant drawings, references, etc. where necessary)	If No or not applicable, please explain
6.1	Drainage Grade all roof areas and open terraces to fall, with adequate gradients, so storm water is directed away from door openings and flow towards rainwater outlets/ floor waste outlets.			
6.2	Slope the RC roof slab and roof gutters at 1:50 or steeper to facilitate efficient run-off.			
6.3	Design two-directional fall to improve drainage and to reduce roof slab thickness.			
6.4	Provide drainage outlet with “debris trap” to prevent choking of the drainage system. Use leaf guards and overflow spouts to prevent water stagnation.			
6.5	Expansion Joints Incorporate adequate expansion joints to prevent cracking and support the vibration of plants and equipment on the roof. The expansion joints should be at regular intervals over large continuous roof areas due to continuous weather exposure.			
6.6	Fixtures of façade access equipment on the roof should not affect/damage the components of the lightning protection system.			

1.7. Building access, spaces, and elements

Objectives:

Ensure that access design and systems enable ease of maintenance to various building elements, vertical spaces, and services.

7.	<u>Access</u>	Y / N / NA	Description of provision (Attach relevant drawings, references, etc. where necessary)	If No or not applicable, please explain
7.1	Air Well and Atrium a) Avoid constricted areas, air wells and atrium which are difficult to access for maintenance. b) Provide suitable access system to maintain the façade/ glazed surfaces within air well/ atrium (e.g. maintenance tracks systems with suspended platforms around large voids and air wells). c) Avoid high volume voids or “soft storeys” with “flying” beams” which do not have easy means of access for maintenance.			
7.2	Ceiling Spaces and Height Avoid lobby areas with high ceiling spaces (e.g., > 5m) which are difficult to access for maintenance.			
7.3	Where ceilings exceed 5m in height, provide catwalks, elevated platforms and/or unobstructed access route for maintenance equipment and vehicles e.g., scissors lift and mobile elevated platforms.			
7.4	Provide a service lift of adequate capacity to transport the maintenance equipment and vehicles to for maintaining high ceiling spaces of sky terraces.			
7.5	Outdoor Lightings Lampposts should be safely and easily accessible for lamp replacement (e.g., areas should have firm surface to allow proper deployment of ladders).			

1.8. Materials and finishes for building access, spaces, and elements

Objectives:

Use appropriate materials and finishes to minimize the frequency of cleaning, repair and replacement.

8.	<u>Materials & Finishes</u>	Y / N / NA	Description of provision (Attach relevant drawings, references, etc. where necessary)	If No or not applicable, please explain
8.1	Walls Avoid spray textured finish on walls and columns where there is heavy pedestrian traffic.			
8.2	Floors Use stain resistant (e.g., mid tone colors) floor finishes in areas with heavy pedestrian traffic such as lift lobbies and common corridors.			
8.3	Avoid combining metal elements with natural stones to reduce risk of staining associated with metal corrosion.			
8.4	Use metal (e.g., aluminum-finished) fixtures or doors (with appropriate fire rating, where necessary) instead of timber in areas exposed to weather or damp conditions.			
8.5	Exterior Ceiling Use weather and wind resistant ceiling materials (e.g., metal panels) in areas exposed to weather elements.			
8.6	External Signage Use weather-resistant materials in semi-gloss finish for easy maintenance.			
8.7	Vehicle Drop-off Areas For vehicle drop-off areas, use oil-resistant surface (e.g. pavement or stone) to reduce permanent staining caused by vehicles. Avoid: porous, soft stone materials, glossy surfaces, or light-colored materials, in areas with heavy pedestrian traffic.			

8.	<u>Materials & Finishes</u>	Y / N / NA	Description of provision <i>(Attach relevant drawings, references, etc. where necessary)</i>	If No or not applicable, please explain
8.8	Driveway Paving Avoid the use of rough stone paving for driveway as they are more prone to trapping dirt.			
8.9	Timber All timber finishes should be kiln-dried and well- seasoned to prevent shrinkage, splits and deformation. All timber finishes should be treated for fungi and insect attack, including anti-termite treatment, with an approved wood preservative.			
8.10	Masonry, Stone and Tiles Provide appropriate and compatible sealer/ impregnator to the sides, top and bottom surfaces of stone works.			
8.11	Where masonry/ stone finishes are used near wet areas, the stone should be appropriately treated (e.g., sealant, admixture, water repellent) and detailed to prevent efflorescence.			

1.9. Design and detailing for building access, spaces, and elements

Objectives:

Provide effective design & detailing for protection against weathering and enabling simple maintenance.

9.	<u>Design & Detailing</u>	Y / N / NA	Description of provision <i>(Attach relevant drawings, references, etc. where necessary)</i>	If No or not applicable, please explain
9.1	Linkways and Drop-off Areas Provide sufficient roof gradient (min 3° slope for flat roof) for efficient water run-off.			
9.2	Design of linkways, drop off porches, and external corridors should minimize ingress of rain, ensuring at least 50% dry covered passage, e.g. Provide drop panels and weatherproof louvers for high volume roof, provide sheltered area or canopy with a depth of at least 2 times the entrance width.			
9.3	Design of linkways and drop-off porches should be modularized with demountable parts (no welded parts) for ease of maintenance, where required, and to facilitate access for maintenance vehicle.			
9.4	Provide concrete curb / collar at the base of metal columns to minimize contact with water and reduce corrosion.			
9.5	Floor slabs should come with removable panels for inspection and maintenance of concealed services.			
9.6	Grade all exposed floors and flat surfaces to fall so as to direct run-off to the external drains and scuppers.			
9.7	Avoid corners and cavities which are inaccessible to cleaning machines.			
9.8	Provide anchor points for safety harness on linkways and canopies for maintenance personnel.			

9.	<u>Design & Detailing</u>	Y / N / NA	Description of provision (Attach relevant drawings, references, etc. where necessary)	If No or not applicable, please explain
9.9	Entrance to Building Provide floor mats with integrated drainage system at the main entrance to reduce entry of water, dirt and dust into buildings. Floor mat should flush with the floor finishes to prevent tripping hazards.			
9.10	Ceiling Avoid monolithic ceiling design (joint-less with no access to internal services) which are more costly and tedious to repair and maintain.			
9.11	Use suspended modular ceiling panels that are easily demountable for its easy replacement and maintenance of embedded services.			
9.12	Where metal ceiling panels are used, they should be designed to prevent sagging and withstand wind loads. Panels should be sized such that they can be easily handled by one person.			
9.13	Staircase Provide appropriate nosing's to avoid chipping of edge of the steps.			
9.14	Expansion Joints Ensure that sufficient expansion joints are provided to external and internal finishes to prevent cracking, warping, etc.			
9.15	Drainage All concealed drainage, rainwater down pipes should have access openings.			
9.16	Recessed Floor Lightings Avoid the use of recessed floor lights that is prone to water damage. If such lights are used, provide drainage to prevent waterlogging.			
9.17	Safety of Maintenance Personnel Provide appropriate warning signs for maintenance crew to prevent stepping			

9.	<u>Design & Detailing</u>	Y / N / NA	Description of provision (Attach relevant drawings, references, etc. where necessary)	If No or not applicable, please explain
	on ceiling boards and non- load bearing surfaces (e.g., in voids of service risers).			

1.10. Parking area access design

Objectives:

Provide adequate and unobstructed access for maintenance personnel and equipment.

10.	<u>Access</u>	Y / N / NA	Description of provision (Attach relevant drawings, references, etc. where necessary)	If No or not applicable, please explain
10.1	Where services/ plant rooms are within car park, provide service route(s) with adequate headroom and sufficient working space for maintenance vehicle/ equipment. (e.g., access and maneuvering space for refuse collection truck).			
10.2	Car Park lots should not block the doors of M&E rooms.			

1.11. Materials and finishes for parking areas

Objectives:

Selection of materials and finishes should take into consideration stains and wear and tear due to vehicular traffic and exhaust fumes.

11.	<u>Materials & Finishes</u>	Y / N / NA	Description of provision (Attach relevant drawings, references, etc. where necessary)	If No or not applicable, please explain
11.1	Provide epoxy coating on car park driveway and parking lot for greater durability. Use resilient material (e.g., thermoplastics) for directional and parking lot markings.			
11.2	The surface material of the driveway and walls should allow easy removal of oil and water stain as well as heavy-duty washing.			

1.12. Design and detailing parking areas

Objectives:

Provide effective design & detailing to protect against damages from vehicles and ensure optimal surface run-off to avoid water stagnation.

12.	<u>Design & Detailing</u>	Y / N / NA	Description of provision (Attach relevant drawings, references, etc. where necessary)	If No or not applicable, please explain
12.1	Guards and Wheel Stoppers Use wheel stopper and crash guards to protect columns, walls, and exposed pipes and services near car park lots and driveways.			
12.2	Ramps Provide groove lines on access ramps to facilitate water discharge to the scupper drains on the side of ramps.			
12.3	Drainage Provide adequate drain outlets to prevent water stagnation (e.g., scupper drains in the vicinity of wheel stoppers).			

12.	<u>Design & Detailing</u>	Y / N / NA	Description of provision <i>(Attach relevant drawings, references, etc. where necessary)</i>	If No or not applicable, please explain
12.4	Provide cut-off drains with adequate depth and gradient at all areas of the car park that are exposed to weather including: the top and bottom of ramps. open/partially exposed staircase (at intermediate landing and/or base of staircase).			

1.13. Materials and finishes for M&E rooms

Objectives:

Selection of materials and finishes for the floors and walls should take into consideration wear and tear and exposure to weather elements.

13.	<u>Materials & Finishes</u>	Y / N / NA	Description of provision <i>(Attach relevant drawings, references, etc. where necessary)</i>	If No or not applicable, please explain
13.1	Floors Use darker tone epoxy topcoat for the floor finishes in service rooms and service corridors.			
13.2	Walls Provide metal chequered plate on walls (up to 1.2m high) for surface protection in service rooms and service corridors that are subject to frequent, heavy wheeled traffic.			
13.3	Doors Provide door kick plate for service rooms that are subject to frequent, heavy wheeled traffic.			
13.4	Use corrosion-resistant material (e.g., aluminum doors) for service doors exposed to weather or damp conditions.			

1.14. Design and detailing for M&E rooms

Objectives:

Provide effective design and detailing to prevent water ingress/egress problems and to enable simple maintenance.

14.	<u>Design & Detailing</u>	Y / N / NA	Description of provision <i>(Attach relevant drawings, references, etc. where necessary)</i>	If No or not applicable, please explain
14.1	Avoid locating wet areas above critical service rooms (e.g., server rooms, LAN rooms, and electrical rooms). If this is not possible, provide double slab with access and adequate height between slabs for maintenance.			
14.2	Provide raised kerb at AHU room, water tank room and electrical rooms to prevent water ingress/egress.			
14.3	External AC Screens Design the sound barriers/ visual screens for air- conditioning equipment in easily demountable, modularized units so as to facilitate easy access and maintenance.			
14.4	Louvres Provide at least 300mm overhang above louvre vents facing external or set back the louvre vents by 300mm from the building edge.			

1.15. Washroom access

Objectives:

The design and layout of toilet facilities should allow easy access for maintenance.

15.	<u>Access</u>	Y / N / NA	Description of provision <i>(Attach relevant drawings, references, etc. where necessary)</i>	If No or not applicable, please explain
15.1	Cleaning Machines and Equipment Provide at least one service lift to facilitate the transport and access for cleaning machines and equipment.			
15.2	The washroom entrance should be wide enough for cleaning machines and equipment.			
15.3	Circulation Adopt “doorless” or vestibule entry to facilitate ease of access and improve ventilation.			
15.4	The layout of toilet should minimize circulation space and adequately designed to handle the anticipated peak traffic.			

1.16. Materials and finishes for washroom

Objectives:

Use materials that are resistant to damp and chemical environments.

16.	<u>Materials & Finishes</u>	Y / N / NA	Description of provision <i>(Attach relevant drawings, references, etc. where necessary)</i>	If No or not applicable, please explain
16.1	Use moisture impervious, durable (e.g., ceramic tiles and phenolic panels) and cleanable materials for toilet wall and floor surfaces so as to facilitate cleaning and resource conservation (such as minimizing the use of water and cleaning agents).			
16.2	Provide slip-resistant flooring (e.g., slip resistant index of R11) for safety of maintenance workers.			
16.3	Use metal gratings of sufficient quality (e.g., Stainless Steel 304) to prevent rusting.			

1.17. Design and detailing of washrooms

Objectives:

Provide effective design and detailing to prevent water damages and to facilitate easy cleaning and maintenance.

17.	<u>Detailing and Design</u>	Y / N / NA	Description of provision <i>(Attach relevant drawings, references, etc. where necessary)</i>	If No or not applicable, please explain
17.1	Fixtures Provide wash basins of appropriate size and depth to reduce splashing and wetting of floors.			
17.2	Use modular and smaller basin mirrors instead of full- span wall mirrors for ease of cleaning.			
17.3	Use suspended cubicle partitions to facilitate floor cleaning and prevent partition panels from constant contact with damp floor.			
17.4	Use suitable hinge and lock for cubicle doors considering the material and weight of doors to prevent misalignment after prolonged use.			
17.5	Ensure that the size of recessed/ built-in trash bin can adequately meet the waste capacity demand.			
17.6	Provide a minimum of 25mm level difference between the corridor and the toilet floor.			
17.7	Use wall-mounted wash basins, urinal bowls, and water closets to facilitate floor cleaning.			
17.8	Door frame/architrave should be raised above the wet areas and terminate above the floor level. Consider installing sub-frames above the screed level to reduce the chance of water ingress from the wet floor through the masonry works.			
17.9	Dispensers Install dispensers' (soap/ hand towel) above the vanity top for ease of replacement/refill. Avoid			

17.	<u>Detailing and Design</u>	Y / N / NA	Description of provision <i>(Attach relevant drawings, references, etc. where necessary)</i>	If No or not applicable, please explain
	installing the dispensers behind mirror/ feature.			
17.10	The dispenser should have a visual indicator (e.g., transparent window) to show the amount of soap/ towel in the dispenser to assist in timely refilling.			
17.11	Install dispensers (soap/hand towel) / hand dryer close to wash basin.			
17.12	Waterproofing Apply waterproofing membrane to the wall or substrate immediately adjacent or behind a washbasin, sink or similar fixture at a height of not less than 300mm above the fixture.			
17.13	Waterproofing membrane should be dressed up at pipe penetrations to the finished floor level and dressed down at least 50mm into the floor outlet. Waterproofing membrane should be applied at least 100mm horizontally around the pipe.			
17.14	Avoid laying tiles directly onto the waterproofing membrane. As a protective measure against damaging the membrane during tiling, a layer of screed should be laid over the membrane when it is cured.			
17.15	Shower Facilities Provide raised kerbs of at least 100mm at the base walls to act as barriers against lateral movement of water.			
17.16	Apply waterproofing membrane with an upturn of at least 300mm to create a minimum tanking protection against migration of water to spaces adjacent or below the wet area.			

17.	<u>Detailing and Design</u>	Y / N / NA	Description of provision (Attach relevant drawings, references, etc. where necessary)	If No or not applicable, please explain
17.17	Apply waterproofing membrane of at least 1800mm height and 1500mm width of the wall or the entire enclosure at bath and shower areas.			
17.18	Supporting Facilities for Cleaners Provide a janitor closet/ store on each level for storage of daily-use cleaning tools/equipment, chemicals, and supplies. Storerooms should come with a large sink for filling and emptying cleaning buckets.			
17.19	Provide a designated room of sufficient size for cleaners to rest and store personal belongings.			
17.20	Provide a central storeroom/area for storage of large cleaning machines (e.g., ride-on sweeper/scrubber). Access doors and aisles need to be large enough for the machines' movement.			

2. Mechanical and Electrical

2.1. Access to M&E installations on façade and roof

Objectives:

Provide safe and easy access for the inspection and maintenance of M&E installations.

2.1	<u>Access</u>	Y / N / NA	Description of provision (Attach relevant drawings, references, etc. where necessary)	If No or not applicable, please explain
2.11	Air Conditioning Condenser Unit Provide adequate working space around the condenser unit to accommodate maintenance and replacement.			

2.1	<u>Access</u>	Y / N / NA	Description of provision (Attach relevant drawings, references, etc. where necessary)	If No or not applicable, please explain
2.12	Externally mounted M&E installations Provide proper means of access for maintenance for externally mounted M&E installations: access to LPS installations (strike pads, down conductor tapes etc.) on façade, replacement of lights on the façade as well as light fittings located in high volume spaces (canopy, drop-off porch, sky terrace, etc.), maintenance of other externally mounted M&E installations (e.g., ACMV louver, CCTV, and PA/AV system, etc.).			
2.13	Lifts Service lift should preferably land at roof level. Lift should be designed with adequate capacity to transport equipment and parts for replacement and maintenance purposes.			
2.14	Provide designated access for motor room-less lift (e.g., 'knock-out wall panel') and hoisting beam for servicing and replacement of lift motor.			
2.15	M&E Installations on Roof Provide adequate working space (e.g., 600m) around and between M&E installations at the roof (cooling tower, generator, condenser unit, photovoltaic panels, etc.) to accommodate easy maintenance and replacement.			
2.16	Lightning Protection System Provide adequate means of access for workers to inspect and maintain air terminals and lightning tapes at edge of the building.			

2.2. Materials and finishes for M&E installations on façade and roof

Objectives:

External M&E services and equipment should be able to withstand exposure to weather elements.

2.2	<u>Materials & Finishes</u>	Y / N / NA	Description of provision <i>(Attach relevant drawings, references, etc. where necessary)</i>	If No or not applicable, please explain
2.21	Use weather-resistant material for all external signage with lighting.			
2.22	Use weather-proof material for outdoor M&E devices such as lightings, speakers, cameras, card reader: IP65 rated lightings and relevant accessories (transformers/controllers/drivers etc.), e.g., near green wall since exposed to dirt and irrigation. External access control equipment (cameras, card reader, etc.) should come with weatherproof covers and/or made of weather resistant materials Avoid locating fire alarm call points in weather- exposed areas. Alternatively, call points should be made of weatherproof material.			
2.23	Use corrosion-resistant material for pipes, ducts, louvers, cables, and trays (including their fixing accessories) that are exposed to weather.			
2.24	Use weatherproof materials for M&E services and equipment. Provide suitable cover/shelter to equipment or component (e.g., electrostatic filters for kitchen exhaust, inverters of photovoltaic installations) that do not come with weather-proof option.			

2.3. Design and detailing M&E installations for façade

Objectives:

Provide effective design and detailing to enable proper functioning and ease of maintenance for M&E equipment.

2.3	<u>Design & Detailing</u>	Y / N / NA	Description of provision (Attach relevant drawings, references, etc. where necessary)	If No or not applicable, please explain
2.31	Lightning Protection System Use natural down conductor (e.g., steel in reinforced concrete structures) instead of exposed lightning tape.			
2.32	Avoid mounting aluminum tape directly onto calcareous building surfaces (e.g., concrete limestone and plaster) to minimize aluminum corrosion.			
2.33	External Air-conditioning Units Avoid locating air-conditioning units in recesses or enclosed areas (e.g., basements) to mitigate the rejected heat from being recirculated into the equipment.			
2.34	Lighting on Roof Provide adequate lighting to access routes leading to M&E installations.			
2.35	Cooling towers Provide adequate water points and power points close to cooling towers to facilitate cleaning and top-up in the event of pump failure (serving make-up water tank).			
2.36	Provide effective cooling tower water treatment to prevent corrosion, fouling, and algae growth.			
2.37	Photovoltaic (PV) Panels Provide adequate water points for cleaning of PV panels.			

2.4. Access to M&E installations in common areas, plants, and equipment rooms

Objectives:

Provide safe and easy access to various M&E installations to enable efficient inspection, repair, and replacement.

2.4	<u>Access</u>	Y / N / NA	Description of provision <i>(Attach relevant drawings, references, etc. where necessary)</i>	If No or not applicable, please explain
2.41	Overhead Services Provide adequate maintenance access space to all M&E installations (light fittings, ventilation equipment, fire alarms, ELV equipment etc.) above the suspended ceiling.			
2.42	Provide ceiling access panels near floor traps, valves, water heater and air conditioning equipment to facilitate inspection/ maintenance of services.			
2.43	Provide multi-tier trays with at least 150mm access space in between for installation/ maintenance of services cables.			
2.44	Provide a minimum of 600mm clear depth between beam soffits and false ceilings for insulated chilled water pipes to pass below the beam. Provide a minimum of 400mm clear depth between beam soffits and false ceilings for insulated refrigerant pipes to pass below the beam.			
2.45	Prefabricated MEP System Where prefabricated MEP is used, provide a minimum workspace of 600mm along the center of the modules. The width of services on each side of the access space should be kept within one arm's length to allow easy reach to all the services.			
2.46	Headroom Provide minimum 2m clear headroom in M&E rooms and access walkways/ service corridors to facilitate safe access and ease of maintenance.			

2.4	<u>Access</u>	Y / N / NA	Description of provision <i>(Attach relevant drawings, references, etc. where necessary)</i>	If No or not applicable, please explain
	Obstructions that result in reduced headroom should be clearly identified and marked.			
2.47	High Volume Spaces For easy access, provide signs and location indicators for services located above high ceiling areas.			
2.48	Provide permanent guarded ladders, catwalks, and hydraulic powered platforms for occasional access to high volume spaces. Such maintenance activities should not require dismantling of services and building elements.			
2.49	Provide scaffolding, staging and temporary ladders for infrequent access to high volume spaces. Ensure any fixtures below such spaces can be easily removed to make space for installation of temporary access systems.			
2.4.10	Sanitary and Rainwater downpipes Avoid casting sanitary pipes within wall structures. Sanitary waste stacks and downpipes should be easily accessible from common areas for maintenance.			
2.4.11	'Cleaning eye' for sanitary/waste piping system should be clear from obstruction and easily accessible for maintenance. Avoid placing 'cleaning eyes' at high and inaccessible areas.			
2.4.12	Lightning Protection System Provide adequate means of access to inspect and test the test-link panels and earth pits.			
2.4.13	Working Space Provide at least 600mm unobstructed space or larger (manufacturers' requirements) for maintenance around all M&E equipment for safe maintenance and replacement of components.			

2.4	<u>Access</u>	Y / N / NA	Description of provision (Attach relevant drawings, references, etc. where necessary)	If No or not applicable, please explain
2.4.14	Ensure adequate working space for the installation and removal of mechanical equipment (e.g. 2.5m unobstructed space at the back of chiller for replacement of chiller tube or unobstructed space to operate lifting equipment).			
2.4.15	Plumbing & Sanitary Provide access panel of minimum 600 x 600 mm for plumbing maintenance.			
2.4.16	Provide adequate working space within service shaft to allow access for maintenance.			
2.4.17	Waste Management Provide vehicle access to grease traps.			
2.4.18	Safety for Maintenance Personnel Provide adequate lightings along access walkways and in M&E rooms.			

2.5. Materials and finishes for M&E installations in common areas, plants, and equipment rooms

Objectives:

Use suitable materials that are resistant to deterioration and exposure to weather elements and dampness.

2.5	<u>Materials & Finishes</u>	Y / N / NA	Description of provision (Attach relevant drawings, references, etc. where necessary)	If No or not applicable, please explain
2.5.1	Use durable (e.g., metal clad) lighting switches and power sockets in M&E rooms			
2.5.2	Waterproofing Provide at least IP65 rated power sockets in wet areas (e.g., pantry, water dispenser, toilets, wash area and potential wet areas in M&E rooms).			

2.5	<u>Materials & Finishes</u>	Y / N / NA	Description of provision (Attach relevant drawings, references, etc. where necessary)	If No or not applicable, please explain
2.5.3	Sanitary and Rainwater downpipes Provide anti-corrosion material for piping and installations (e.g., UPVC pipes).			
2.5.4	Fabric Ducting System Use fabric ducting system for large spaces (e.g., atrium and auditorium) for ease of maintenance and replacement.			

2.6. Design and detailing M&E installations in common areas, plants, and equipment rooms

Objectives:

The design and placement of M&E installations should aim at minimizing maintenance interventions and enabling ease of repair and replacement.

2.6	<u>Design & Detailing</u>	Y / N / NA	Description of provision (Attach relevant drawings, references, etc. where necessary)	If No or not applicable, please explain
2.6.1	Provisions for Maintenance Provide switch socket outlets (each point to cover a radius of max. 15m) in common areas to allow use of powered equipment for maintenance.			
2.6.2	Provide sufficient water points and socket points to facilitate the use of electrical cleaning equipment in washrooms. Socket points should be provided with splash proof covers.			
2.6.3	Equip cleaners' storerooms with utilities supply for washing of cleaning tools and equipment, as well as drainage points and splash proof socket points to facilitate the charging of cleaning tools/equipment.			
2.6.4	Condensation Treatment			

2.6	<u>Design & Detailing</u>	Y / N / NA	Description of provision <i>(Attach relevant drawings, references, etc. where necessary)</i>	If No or not applicable, please explain
	Provide condensation treatment to air distribution outlets located in high humid areas.			
2.6.5	Prefabricated MEP System Adopt prefabricated Mechanical, Electrical, and Plumbing (MEP) systems for more productive replacement and installation.			
2.6.6	Labelling Provide labelling and color coding of piping and conduits consistent with industry standards (e.g., provide directional signs and valve status on pipes for ease of identification).			
2.6.7	Provide isolation valves and proper identification to all pipes and ductworks for easy maintenance.			
2.6.8	High Volume Spaces Use wall mounted lights and/or suspended lighting systems for rooms with high ceilings (e.g., atria, lecture theatres, reception or lobby areas and stairwells).			
2.6.9	Parking Areas Locate electrical services away from voids that are exposed to weather elements.			
2.6.10	Avoid locating electrical outlets on columns or walls adjacent to car park lots.			
2.6.11	Provide adequate water and power points at appropriate locations (each point to cover a radius of max. 15m) to facilitate washing and cleaning.			
2.6.12	Sanitary and Rainwater downpipes Avoid locating sanitary and waste pipe in areas with high pedestrian traffic (e.g., above escalators and within high voids above common staircases, food preparation area and water tank).			

2.6	<u>Design & Detailing</u>	Y / N / NA	Description of provision <i>(Attach relevant drawings, references, etc. where necessary)</i>	If No or not applicable, please explain
2.6.13	Avoid concealing drainpipes in the screed of dry areas.			
2.6.14	Manholes Depending on their location, manhole covers should be designed to withstand the required loads. Manholes should be located away from high traffic areas to minimize risk to workers during maintenance.			
2.6.15	Where possible, manhole cover / cable trench cover/ grating should be moveable by one person. Where covers are too heavy to be lifted by one person, lifting tools must be provided.			
2.6.16	Bin Centre Provide water point and discharge point inside the bin center for regular cleaning purposes.			
2.6.17	Lighting and Ventilation Provide sufficient lighting and ventilation in M&E rooms and risers that houses heat emitting equipment (e.g., electrical panels, IT switches). Provide backup emergency lighting in event of power failure.			
2.6.18	Flood Mitigation Provide flood mitigation measures (such as sump pump) for M&E rooms located at basements.			
2.6.19	Ductworks/ Pipes across Floor Avoid ductworks/ pipe installations onto M&E room floors to minimize tripping hazards, and damage due to stepping.			
2.6.20	Redundancy Systems Shutdown maintenance of M&E installations should not affect building's daily operation. Provide redundancy systems and emergency power for shutdown maintenance.			

2.6	<u>Design & Detailing</u>	Y / N / NA	Description of provision <i>(Attach relevant drawings, references, etc. where necessary)</i>	If No or not applicable, please explain
2.6.21	Provide at least 2 lifts for redundancy to avoid downtime during maintenance.			
2.6.22	Lift Motor Room Install lighting switches near entrances of lift motor rooms, preferably within an arm's length, so that the working space can be lit before entering.			
2.6.23	Provide a conducive environment for maintenance personnel in the lift motor room (e.g., well ventilated, provision of fans).			
2.6.24	Provide a minimum 1m overhang above lift motor room doors to prevent rainwater ingress.			
2.6.25	Lift Well Provide 2-way switch for lift well lighting between the lift well and the lift machine room.			
2.6.26	Bulky MEP Equipment Provide removable facade with hoisting platform for replacement of bulky MEP equipment if it cannot be transported via service lift.			
2.6.27	AHU Provide kerb up washing area with water point and floor trap in AHU rooms for maintenance of the filters. Proper mechanical ventilation in the AHU rooms should be provided for maintenance personnel.			
2.6.28	Locate water points at least 1m away from any electrical installations (e.g., control panel) in AHU rooms.			
2.6.29	Direct drainage pipe from AHU into the floor trap instead of terminating the pipe above the floor trap to avoid spillage.			
2.6.30	Avoid placing AHU/FCU directly above critical equipment (e.g., server rack, electrical panel, etc.).			

2.6	<u>Design & Detailing</u>	Y / N / NA	Description of provision <i>(Attach relevant drawings, references, etc. where necessary)</i>	If No or not applicable, please explain
2.6.31	Provide auxiliary insulated drain pan with insulated drainpipe for ceiling mounted FCU in M&E rooms. The auxiliary installation must not affect proper operation of the FCU (e.g., hindrance to airflow of FCU).			
2.6.32	Provide dampers/ vibration isolators to equipment prone to vibration transmittance (e.g., ducted FCU)			
2.6.33	Provide suitable waterproofing for AHU rooms.			
2.6.34	Chiller Provide auto-tube cleaning system for condenser tube to avoid fouling.			
2.6.35	Service Risers Design service risers with adequate width and depth to allow ease of maintenance. (e.g. Minimum width of 600mm should be provided for access to services and components)			
2.6.36	Provide handles (recessed typed) for service riser doors			
2.6.37	Provide riser doors that can be opened from inside without the need of keys.			
2.6.38	Provide load bearing floor in risers that require access by maintenance personnel.			
2.6.39	Uninterruptible Power Supply (UPS) Provide by-pass for maintenance of UPS to prevent downtime.			

2.7. Facility Management

2.7	<u>Digital Readiness and Automation</u>	Y / N / NA	Description of provision (Attach relevant drawings, references, etc. where necessary)	If No or not applicable, please explain
2.7.1	Building Management System (BMS) Install appropriate computer-based BMS to control and monitor M&E equipment such as centralized air-conditioning system, ventilation, lighting, power system, fire protection services, plumbing, security system and landscape irrigation system.			
2.7.2	BMS should cater to specific operational needs of building instead of comprising generic specifications from the industry.			
2.7.3	Use BMS systems which have analytic features such as monitoring, charting, and trending reports for optimal maintenance planning.			
2.7.4	Include alarm notification options (such as SMS) for buildings not operating 24/7.			
2.7.5	Provide remote web-based monitoring.			
2.7.6	Provide common (BACnet, Modbus etc.) or non-proprietary protocol for the network backbone of the building management system (BMS). (This will enable standard compliant sensors or equipment to be easily added to the building and be monitored through the BMS)			
2.7.7	Use common file formats for automated and scheduled exported data from any chosen data points. (This will enable exchange of data between systems and allow such data to be analyzed by analytics or optimization software readily).			
2.7.8	Sensors Place sensors strategically with consideration to access, accuracy and operation.			

2.7	<u>Digital Readiness and Automation</u>	Y / N / NA	Description of provision <i>(Attach relevant drawings, references, etc. where necessary)</i>	If No or not applicable, please explain
2.7.9	Sub-metering Divide sub-metering by energy loads and areas. (This will facilitate energy audit, identify maintenance needs and targets to reduce energy use).			
2.7.10	Building Information Modelling (BIM) Use BIM to facilitate integration of relevant information on facility and asset management and better coordinated documentation for maintenance operations.			
2.7.11	Adopt comprehensive digital building management through 3D building model integrated with sensors, video analytics and command center. (e.g., facial recognition, incident reporting, and crowd monitoring etc.)			
2.7.12	Documentation Prepare adequate documentation to ensure operational continuity from design to maintenance, comprising: Written description of all the systems operation and control actions to meet the functional requirements of the system, including control strategy and logic diagrams. Integration test results of air and hydronic systems and gases (CO & CO2). Network schematic of the BMS network, showing device locations and network addresses. Description of BMS point list including user adjustable points (IT/BMS), hard and derived points, and their respective controller and register addresses. Instruction's manuals and training guides and quick reference procedures/ checklist			
2.7.13	Power and Data Provision			

2.7	<u>Digital Readiness and Automation</u>	Y / N / NA	Description of provision <i>(Attach relevant drawings, references, etc. where necessary)</i>	If No or not applicable, please explain
	Provide adequate power and data provisions (LAN cables, Wi-Fi, network switches, server with spare capacity etc.) for future smart FM features.			

2.8. Security

2.8	<u>Access and Control</u>	Y / N / NA	Description of provision <i>(Attach relevant drawings, references, etc. where necessary)</i>	If No or not applicable, please explain
2.8.1	Locate central control console in areas with 24-hours manning.			
2.8.2	Locate the fire command center (FCC) and security center to oversee at least one main ingress/ egress point or critical area. Design the FCC such that the operator is able to view all alarm signage from his sitting position instead of having to turn round to see the alarm soundings.			
2.8.3	Use turnstile with proximity security access passes.			
2.8.4	Remote monitoring/ CCTV Use CCTV surveillance technology with motion sensing alert to reduce security personnel. Provide CCTV surveillance at all common and critical areas including but not limited to: Car Park Lift lobbies Entries into staircases Main lobbies All exits of the building/ car park Last landing of escape staircase staircases Loading/unloading area Essential plants and equipment rooms			
2.8.5	Visitors Management System Use computer-based Visitors Management System to record usage of facilities by specific visitors			

2.8	<u>Access and Control</u>	Y / N / NA	Description of provision <i>(Attach relevant drawings, references, etc. where necessary)</i>	If No or not applicable, please explain
	and provide documentation of visitor's whereabouts.			
2.8.6	Natural Surveillance Natural surveillance of a concealed or isolated route should be encouraged. For example, provide external- facing windows or openings for enclosed stairs or ramps such that they are visible from the surrounding properties.			

3. Landscape

3.1. Landscape access and maintenance

Objectives:

Provide appropriate access to planters, green walls, and other landscaped features to carry out pruning, maintenance, and replacement safely and efficiently.

3.1	<u>Access</u>	Y / N / NA	Description of provision <i>(Attach relevant drawings, references, etc. where necessary)</i>	If No or not applicable, please explain
3.1.1	Access Route Landscaped areas should have suitable access routes (e.g., pavement, gravel, mulch paths) of adequate width to facilitate maintenance access for workers, equipment, and materials.			
3.1.2	Vertical Greenery Frontal access to green walls (including its sub-system – such as substrate, drainage, irrigation pipelines, supporting sub-frames, gratings, etc.) For green walls that are less than 2m in height, allow maintenance access from the front via elevated work platforms (e.g. ladder stand platform, mobile tower scaffold, and pole pruner).			

3.1	<u>Access</u>	Y / N / NA	Description of provision <i>(Attach relevant drawings, references, etc. where necessary)</i>	If No or not applicable, please explain
3.1.3	For green wall that are more than 2m in height, provide an unobstructed, flat, stable surface of adequate structural capacity in front to allow safe and effective deployment of MEWPs (mobile elevated work platforms). Where the foreground is turfed, the latter should comprise supporting underlayers to withstand the load of MEWPs.			
3.1.4	Rear access to green wall (including its sub-system – such as substrate, drainage, irrigation pipelines, supporting sub-frames, gratings, etc.) Provide permanent rear access (e.g., walkways and platforms) with minimally 600mm clear width to all parts of the green wall. The rear access must be designed with edge protection to mitigate risks of fall-from-height.			
3.1.5	Mark and identify all designated access/inspection points for maintaining vertical greeneries.			
3.1.6	Growth lights Provide direct access to growth lights so as not to damage the greenery.			
3.1.7	Sky Garden Provide adequate access and workspace around trees and tall plants.			
3.1.8	Large Turfed Surfaces/ Fields Locate large, turfed surfaces near fire engine access way (with gates of adequate width, if fenced) to facilitate access of equipment e.g., turf mower.			

3.1	<u>Access</u>	Y / N / NA	Description of provision (Attach relevant drawings, references, etc. where necessary)	If No or not applicable, please explain
3.1.9	Irrigation Provide adequate access to irrigation systems for maintenance and replacement. This includes subsurface irrigation systems.			
3.1.10	Provide proper access (e.g., pavers, gravel or mulch paths) to sources of water supply.			
3.1.11	Water Features Provide safe and direct means of access for maintenance of water features (e.g., lightings in water feature).			
3.1.12	Provide adequate and safe access to pump room, balancing tanks and other pump equipment which serves the water features.			
3.1.13	Storage Provide adequate storage areas for landscape maintenance equipment and materials.			

3.2. Materials for landscape

Objectives:

Select appropriate plants species and landscape materials to minimize the frequency of maintenance interventions such as pruning, cleaning and replacement.

3.2	<u>Materials</u>	Y / N / NA	Description of provision (Attach relevant drawings, references, etc. where necessary)	If No or not applicable, please explain
3.2.1	Plant selection Select plant species in response to the expected environmental conditions (e.g., choose plants which are appropriate for sunny/shady areas, waterlogged grounds, high pedestrian footfall areas, elevated edges, sloped terrain, wind conditions etc.).			

3.2	<u>Materials</u>	Y / N / NA	Description of provision (Attach relevant drawings, references, etc. where necessary)	If No or not applicable, please explain
	Avoid planting fruit trees next to walkways to reduce maintenance due to falling fruits.			
3.2.2	For vegetation abutting water features and water bodies, avoid selecting plant species with excessive shedding of leaves and blooms. Accumulated plant debris may end up choking the water systems.			
3.2.3	Avoid species with invasive roots system near non- suspended pavement, or for sky gardens and roof-top landscape. Provide root barriers in these areas to prevent them from penetrating into structures, weep holes, drains, and floor traps.			
3.2.4	In areas where plant inspection is infrequent and/or cannot be effectively conducted, avoid plants species (e.g., species of bromeliad, alocasia) with thick axils that trap water and require regular monitoring for signs of mosquito breeding.			
3.2.5	Plant ferns and hardy ground covers (instead of turf) on sloping grounds to prevent soil erosion and minimize the need of frequent maintenance.			
3.2.6	Use synthetic turf in areas with high pedestrian traffic or intensive activities (e.g., school fields), as well as areas with difficulty of access (e.g., green islands within pool/ water feature).			

3.2	<u>Materials</u>	Y / N / NA	Description of provision (Attach relevant drawings, references, etc. where necessary)	If No or not applicable, please explain
3.2.7	Test Plots Prior to skyrise greenery installation, set up plant test plots (mock-up surfaces, etc.) in consultation with skyrise greenery consultants to ascertain the horticulture performance of selected greenery systems and species. The test plots will help to identify suitable species as well as inform building owners/ users on the maintenance needs and strategies.			
3.2.8	Water Features Use concealed bracket systems for stone claddings within water features to minimize the occurrence of efflorescence and stains on grouting.			
3.2.9	Avoid water features with rough edges and surfaces as they are prone to dirt accumulation and algae growth.			
3.2.10	Provide backwash for filters; Coarse screens shall be provided to keep dry leaves and rubbish getting into pump. Precautions to be taken in design to avoid pump overheating due to pump suction being choked with debris.			
3.2.11	Provide adequate lighting and ventilation to pump rooms.			

3.3. Design and detailing landscape features

Objectives:

Provide effective design and detailing to facilitate water drainage as well as to enable safe and ease of maintenance.

3.3	<u>Design & Detailing</u>	Y / N / NA	Description of provision <i>(Attach relevant drawings, references, etc. where necessary)</i>	If No or not applicable, please explain
3.3.1	Locate plants with similar water requirement within the same vicinity.			
3.3.2	Planters Provide sufficient soil depth for roots to grow. In general, groundcovers and shrubs require soil depths of between 300mm and 500mm, while small and medium trees (mature heights of 8m to 10m) require between 1m and 1.5m soil depth.			
3.3.3	Planting on Ground Planting should be carried out directly on ground wherever possible to reduce the use of planter boxes. If located on true ground, the planter boxes should be built on true ground and not on concrete surface to ensure proper drainage.			
3.3.4	Un-edged Planting Areas Planting soil should preferably be 50mm lower than unedged planters flushed in level to adjacent hardscape finishes, to reduce soil erosion and spillage onto common and public areas. These are applicable to areas such as true-ground planting, sunken e-deck, roof gardens.			
3.3.5	Drainage Landscaped footpaths should be at least 50mm higher than the drain for effective drainage. The footpaths should be sloped to fall effectively to the nearest drain.			
3.3.6	Ensure runoff from roofs does not fall directly onto landscape areas. Otherwise, provide materials to reduce impact (e.g., gravel).			
3.3.7	Provide appropriate drainage outlets for vertical greenery to avoid stains on façade or water ponding at the base of the installation (e.g., use drainage trays at the base of the installation).			

3.3	<u>Design & Detailing</u>	Y / N / NA	Description of provision <i>(Attach relevant drawings, references, etc. where necessary)</i>	If No or not applicable, please explain
3.3.8	Turfs Avoid planting turf right up to the base of trees to minimize the risk of mechanical damage to tree barks during turf mowing.			
3.3.9	Avoid having isolated thin strips of turf or small grass patches that require frequent mowing.			
3.3.10	Grass surface should be at the same level or slightly higher than adjacent hard surfaces (e.g., pavements) to ease turf mowing.			
3.3.11	Provide buffer strips (e.g., gravel) between the turfed surfaces and pavements/footpaths to prevent soil erosion and spillage. Sub-soil drainage should be avoided.			
3.3.12	Sky Gardens Allow adequate exposure of plants to direct sunlight to avoid etiolation or leaning of trees beyond the edge of building in search of light.			
3.3.13	Where landscape is located near building edges, provide edge protection (e.g., guardrails, lifelines or anchor points) to mitigate risk of fall-from-heights.			
3.3.14	Trees should be adequately secured to withstand strong wind and mitigate risks of uprooting.			
3.3.15	Avoid planting shrubs beyond parapets so that maintenance personnel do not have to lean over the parapet to reach the foliage.			
3.3.16	The floor slab should be sloped towards the drainage outlet with silt control system. Waterproofing membrane should be properly installed around the drain opening so that water drains off from the waterproofing membrane to the roof outlet. Plants should not be allowed to grow into guttering.			

3.3	<u>Design & Detailing</u>	Y / N / NA	Description of provision <i>(Attach relevant drawings, references, etc. where necessary)</i>	If No or not applicable, please explain
3.3.17	Provide adequate sources of water supply (e.g., taps) that are separated from the irrigation system in landscape areas.			
3.3.18	The finished level of sky gardens should be lower than adjoining indoor threshold to avoid rainwater ingress.			
3.3.19	In circumstances where sky gardens finished level is higher than the indoor finished level (e.g., retrofitted roof garden in an existing building), provide up-stand or cut-off drains with adequate capacity to minimize rainwater ingress.			
3.3.20	Provide adequate distance between tall plants and lamp posts to avoid damage to the lamp post by the roots.			
3.3.21	Waterproofing Provide adequate waterproofing system at landscaped areas and planters. Waterproofing should make to be root resistant and/or alongside a suitable root barrier.			
3.3.22	Tree Wells & Gratings Tree well gratings are to be durable and weather- resistant with effective drainage to avoid waterlogging issues. Gratings should be designed to prevent litter from entering the tree well, facilitate inspection access (i.e. designed with moveable modular segments, etc.) to the tree well and cater for installation of tree support system and future tree growth in girth.			
3.3.23	Irrigation system Adopt automatic-irrigation systems with rain sensors.			

3.3	<u>Design & Detailing</u>	Y / N / NA	Description of provision <i>(Attach relevant drawings, references, etc. where necessary)</i>	If No or not applicable, please explain
	Ensure that landscape under sheltered areas remain irrigated even on rainy days.			
3.3.24	Where water supply to the irrigation system comes from rain harvesting systems, effective strainers should be provided to prevent blockages and frequent maintenance to the irrigation network.			
3.3.25	Water Features Use concealed bracket systems for stone claddings within water features to minimize the occurrence of efflorescence and stains on grouting.			
3.3.26	Avoid water features with rough edges and surfaces as they are prone to dirt accumulation and algae growth.			
3.3.27	Provide backwash for filters; Coarse screens shall be provided to keep dry leaves and rubbish getting into pump. Precautions to be taken in design to avoid pump overheating due to pump suction being choked with debris.			
3.3.28	Provide adequate lighting and ventilation to pump rooms.			

4. Other good practices

4.1. Design collaboration

4.1	Design Collaboration	Y / N / NA	Description of provision <i>(Attach relevant drawings, references, etc. where necessary)</i>	If No or not applicable, please explain
4.1.1	Design Brief Provide a clear design brief for maintenance performance and operations.			

4.1	Design Collaboration	Y / N / NA	Description of provision <i>(Attach relevant drawings, references, etc. where necessary)</i>	If No or not applicable, please explain
4.1.2	Collaborate with FM and other stakeholders Engage FM practitioners throughout the planning and design phase for their inputs and review of design drawings and specifications.			
4.1.3	Collaborate with relevant WSH consultants to review the necessary safety provisions for all areas that require maintenance access.			
4.1.4	Virtual Reality Conduct virtual walkthrough of the digital building model to identify potential maintenance issue during design.			

4.2. Documentation

4.2.	Building Records	Y / N / NA	Description of provision <i>(Attach relevant drawings, references, etc. where necessary)</i>	If No or not applicable, please explain
4.2.1	Prepare as-built drawings (showing equipment layout, routing of M&E services, concealed services and landscape) and O&M manuals for end-users' future reference and maintenance purposes. These documentations should be regularly updated and kept for record purposes. All maintenance personnel should be made aware of such records. The records are preferably in both hard and soft copies, with proper dates of updating.			

4.3. Innovation

4.3	Innovation	Y / N / NA	Description of provision (Attach relevant drawings, references, etc. where necessary)	If No or not applicable, please explain
4.3.1	Self-cleansing Facade Adopt façade materials with self-cleansing properties (e.g., titanium dioxide coating) to reduce the frequency of façade cleaning.			
4.3.2	Pneumatic Waste Conveyance System (PWCS) Adopt PWCS to improve productivity in refuse collection and disposal. PWCS can be fitted with sensors to monitor waste disposal patterns and volume. The frequency of waste collection can thus be adjusted accordingly. Provide adequate openings to access PWCS piping's, especially when concealed underground.			
4.3.3	Smart Lighting/Sensor Adopt lighting with sensors to help understand human traffic patterns and optimize/reduce the provision of lighting in low traffic areas. This can help to reduce energy and maintenance.			
4.3.4	Smart Bin Adopt litter/ compactor bins with sensors to facilitate on demand refuse collector			
4.3.5	Smart Toilet Adopt autonomous systems with the use of sensors. Sensors can detect, monitor, and feedback any defects or situations (e.g., ammonia, people traffic, lighting levels, abnormal water usage, hand soap, hand towels, toilet paper and litter bins). Data gathered from sensors can use for analytics such as the determination of peak and off-peak usage and forecast cleaning regimes to optimize cleaning crew deployment.			
4.3.6	Use tiles, basins, urinals, and WCs that are infused/coated with anti-stain material or self-cleaning protective coatings (e.g.,			

4.3	Innovation	Y / N / NA	Description of provision (Attach relevant drawings, references, etc. where necessary)	If No or not applicable, please explain
	fluoropolymer coating) to facilitate cleaning efforts and eliminate the problem of tough stains and graffiti.			
4.3.7	Provide anti-odor tiles that are able to address foul smell such as urine spillage on the floor.			
4.3.8	Digital Pest Control Adopt smart surveillance/ monitoring systems to detect pests (e.g. rodent tracker)			
4.3.9	Autonomous Cleaning Adopt autonomous cleaning robots. Minimize level changes such as ramps and steps to enable its deployment.			
4.3.10	Mobile Incident Reporting Management Adopt mobile app-based incident reporting to facilitate efficient maintenance.			
4.3.11	Lift Sensors Monitoring Adopt autonomous systems with the use of sensors. Sensors can detect, monitor and feedback any predictive events and potential breakdowns (e.g. mislevelling of lift car, abnormal travelling speed of lift, abnormal performance of lift door, and condition of lift rope). Data gathered from sensors can be used for analytics such as determination of usages of lifts and forecast the necessary maintenance regimes. Lift mechanics can be dispatched effectively to rectify the abnormality, and prevent any faulty lift incidents). This can also be extended to escalators			

Appendix-6.0 Project Handover Checklist & Client Acceptance Document [Sample]

Project Name	
Project Ref:	
Prepared By	
Revision	
Date	

Handover					
SN	Item	Responsible for Acceptance	Work acceptable for sign off	Outstanding Items	Notes
1	Building fabric and construction				
1.1	Structure				
1.2	Fabric				
1.3	Fittings				
1.4	Windows				
1.5	Other				

SN	Item	Responsible for Acceptance	Work acceptable for sign off	Outstanding Items	Notes
2	Mechanical Services				
2.1	Gas Installations				
2.2	Heating				
2.3	Water distribution / LA				
2.4	Mechanical ventilation and Air Conditioning Plant				
2.5	Lifts				
2.6	BMS & AMR				
2.7	Labelling				
2.8	Service Contracts				
2.9	Other				

SN	Item	Responsible for Acceptance	Work acceptable for sign off	Outstanding Items	Notes
3	Electrical Services				
3.1	Electrical Cabling				
3.2	Lighting				
3.3	Emergency Lighting				
3.4	Data Cabling				
3.5	Wireless Data				
3.6	AV				
3.7	CCTV				
3.8	Intruder Alarm				
3.9	Fire Alarm				
3.10	Access Control				
3.11	BMS				
3.12	Labelling				
3.13	Service Contracts				
3.14	Lightning Protection				
3.15	Other				

SN	Item	Responsible for Acceptance	Work acceptable for sign off	Outstanding Items	Notes
4	Fire Safety				
4.1	Fire Strategy				
4.2	Fire Alarm				
4.3	Fire Appliances				
4.4	Passive Fire Protection				
4.5	Ironmongery				
4.6	Service Contracts				
4.7	Other				

SN	Item	Responsible for Acceptance	Work acceptable for sign off	Outstanding Items	Notes
5	Security				
5.1	Intruder Alarm				
5.2	CCTV				
5.3	Key Schedule				
5.4	Services Contracts				
5.5	Other				

Building Operation Manual and Health and Safety File Checklist			
BASIC INFORMATION		TEST CERTIFICATES	
General Description of the works		PHE installations	
Contract Details		Electrical Installations – Internal	
Designers (Architects / Engineers etc.)		Electrical Installations – External	
Main Contractor and specialist Sub-contractors		Gas Installations – Internal	
Directory of manufacturers and key suppliers		Gas Installations – External	
CERTIFICATES AND APPROVALS		Cranes, hoists, lifting beams etc.	
Planning Approval		Lifts (passenger & goods)	
Building Regulations Compliance Certificates		Safety ring bolts	
Water Bylaws Compliance Certificate		Boilers & pressure vessels	
NRA Discharge Consents		Mechanical ventilation or Air Conditioning plant	
Structural Calculations Certificate		Internal heating installations	
Hand-over Certificate		External Heating Mains	
Completion Certificate		Other plant/installations as appropriate	
BUILDING FABRIC AND CONSTRUCTION		FIRE, HEALTH & SAFETY	
Structural Data Summary sheets		Schedule of Fire Appliances	
Schedule of Principal Materials		Schedule of Fire risks	
Notice of Restrictions on Use		Schedule of potential hazards	
Schedule of as-built drawings and specifications		Schedule of hazardous substances (if Applicable)	
Schedule of calculations		Updated asbestos information (If Applicable)	
Schedule of information not included in H&S file		Fire Officer's Report	
Copies of documents scheduled above		Statement of fire/safety compliance	
SERVICES, PLANT AND EQUIPMENT		MISCELLANEOUS	
Schedule of as-installed drawings and specifications		Schedule of keys and security devices	
Schedule of information not included in H&S file		Waste disposal records	
Copies of documents as scheduled above			
Schedule of Plant and Equipment			

Literature, maintenance and operating instructions	
Spares list	
Planned Maintenance Documents	
Building Regulations relaxation's or dispensations	

Confirmation of
Acceptance by Client

Signed:	Position:
Name:	On behalf of:
Date:	

Signed:	Position:
Name:	On behalf of:
Date:	

Signed:	Position:
Name:	On behalf of:
Date:	